Discharge calculation from fluorometer experiments

V1: **fixed calibration amounts (50 ml)**

Work through example on

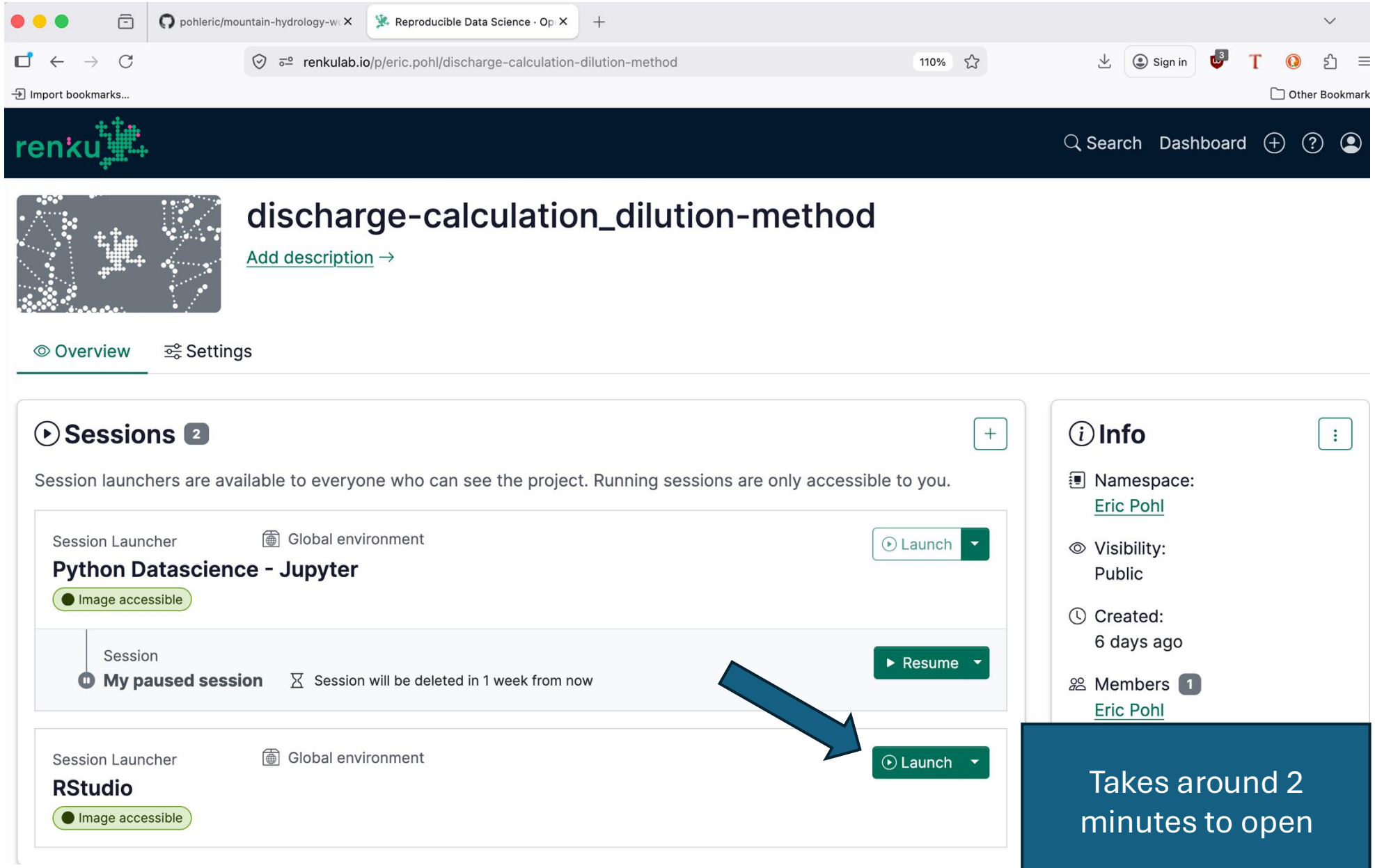
https://renkulab.io/p/eric.pohl/discharge-calculation-dilution-method

In RStudio on renkulab.io (same workflow on your own computer)

Works best on: **Google Chrome**, **Safari**

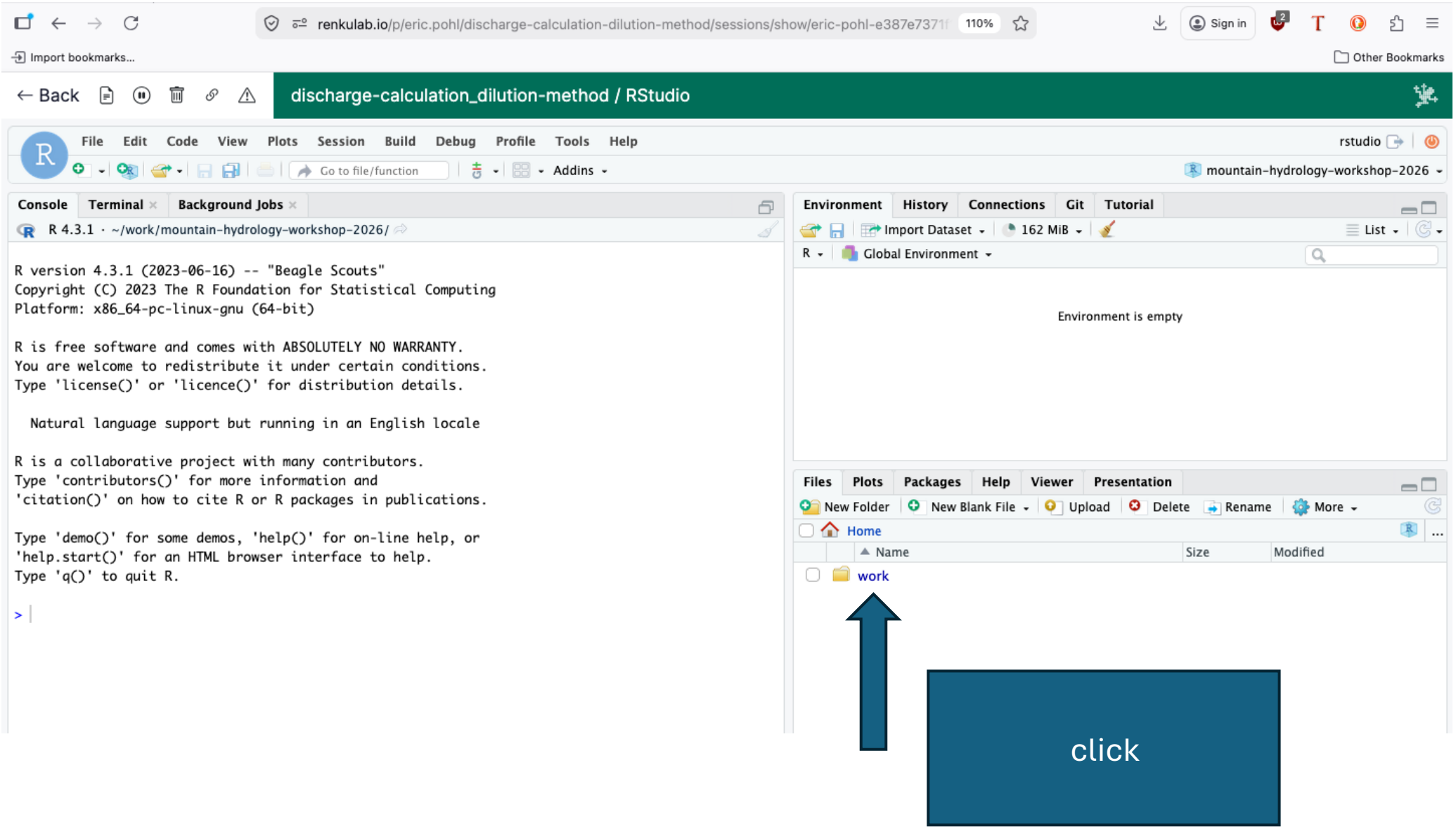
Some problems on: Firefox

https://renkulab.io/p/eric.pohl/discharge-calculation-dilution-method

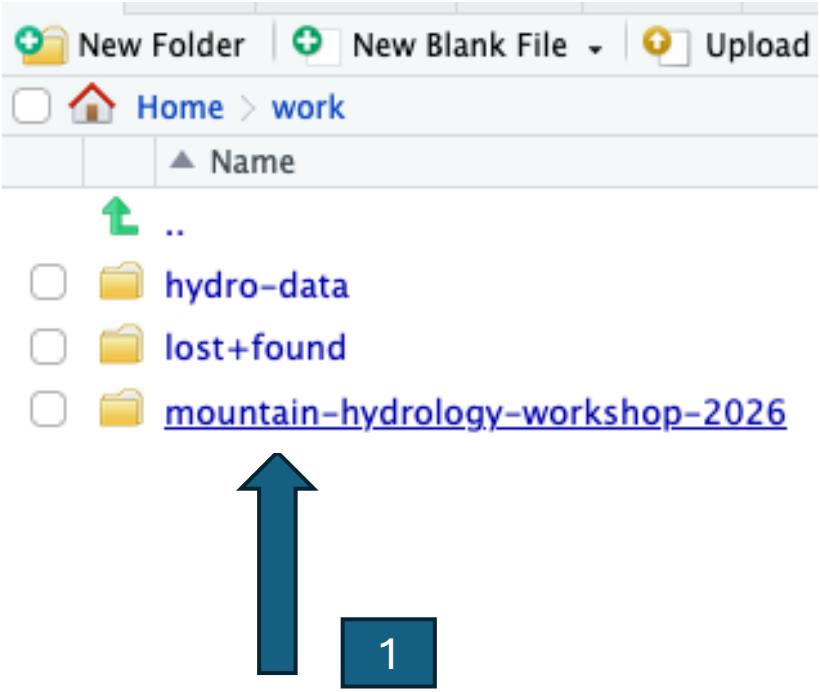


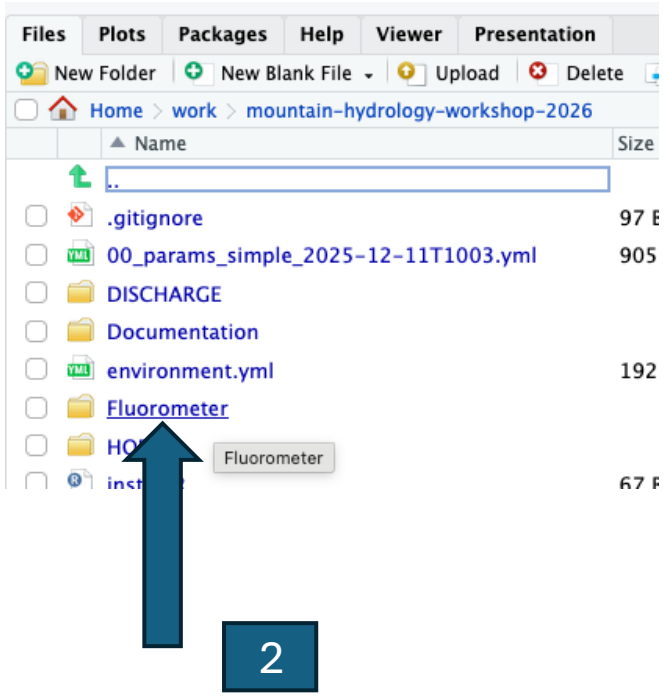
Takes around 2 minutes to open

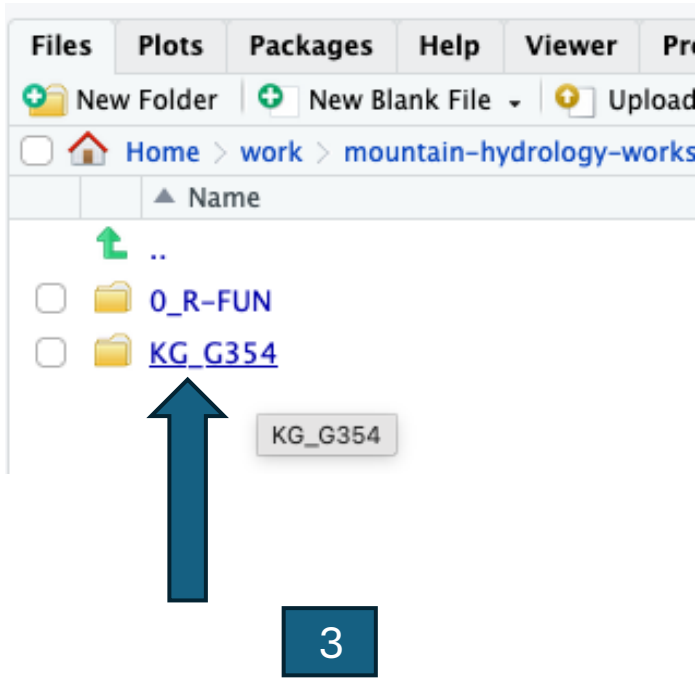
Opening the program. Directory: **work/CA-hydroworkshop/materials/Fluorometer/KG_G354/R/003_Qcalc.R**

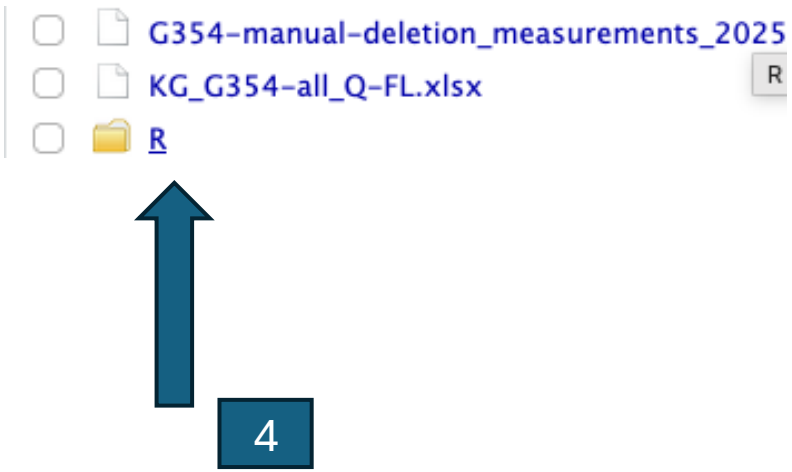


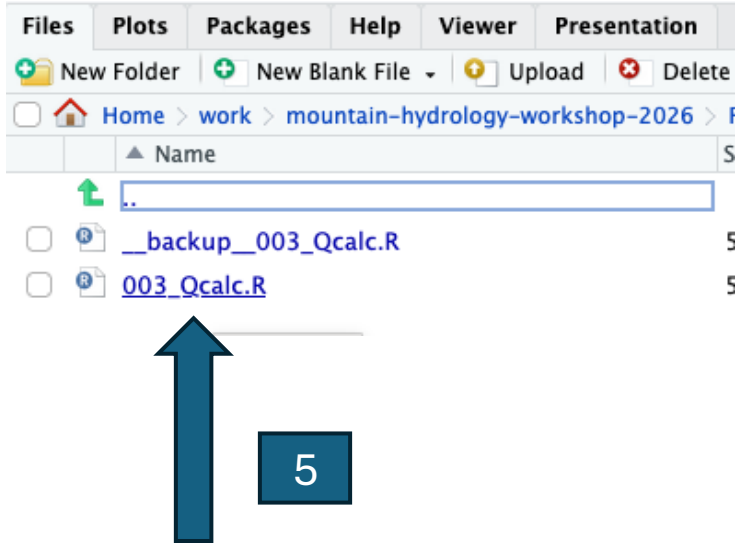
Opening the program. Directory: **work/CA-hydroworkshop/materials/Fluorometer/KG_G354/R/003_Qcalc.R**











Click, click, click ...

Click: “Install” if you see this message

← Back | discharge-calculation_dilution-

File Edit Code View Plots Session Build D Profile Tools Help | rstudio

Go to file/function | Addins | Project: (None)

003_Qcalc.R

Source on Save | Run | Source

Packages MESS and zoo required but are not installed. Install Don't Show Again

```
1  require(readxl)
2  require(lubridate)
3  require(MESS)
4
5  # - reading FL30 raw data
6  # - transforming data into readable format with signal in mV
7
8  # Arguments
9  path = "~/work/mountain-hydrology-workshop-2026/Fluorometer/KG_G354"
10
11 Standard = 50        # g/l - The concentration of the color tracer (Standard solution)
12 TracerVolume = 0.1  # l - How many liters! were injected into the river?
13
14 # ---- 2021 -----
15 # Input file (Excel)
16 filename_excel = '2021 - Measurements_G354.xlsx'
17
18 # ----
```

1:1 (Top Level) R Script

Console Terminal Background Jobs

R 4.3.1 · ~/

```
R version 4.3.1 (2023-06-16) -- "Beagle Scouts"
Copyright (C) 2023 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.
```

Environment History Connections Tutorial

Import Dataset | 158 MiB | List

R | Global Environment

Environment is empty

Files Plots Packages Help Viewer Presentation

New Folder | New Blank File | Upload | Delete | Rename | More

Home > work > mountain-hydrology-workshop-2026 > Fluorometer > KG_G354 > R

Name	Size	Modified
..		
003_Qcalc.R	5.1 KB	Apr 8, 2026, 11:11 PM

This is a “script” (program)

003_Qcalc.R

Source on Save | Run | Source

```r
require(readxl)
require(lubridate)
require(MESS)

# - reading FL30 raw data
# - transforming data into readable format with signal in mV

# Arguments
path = "~/work/mountain-hydrology-workshop-2026/Fluorometer/KG_G354"

Standard = 50        # g/l - The concentration of the color tracer (Standard solution)
TracerVolume = 0.1  # l - How many liters! were injected into the river?

# ---- 2021 -----
# Input file (Excel)
filename_excel = '2021 - Measurements_G354.xlsx'

# ----
sheet_name = '20210928_1052'    # Use this for all measurements
TracerVolume = .02
sheet_name = '20210928_1220_cleaned'    # Use this for all measurements
TracerVolume = .025
sheet_name = '20210928_1310'    # Use this for all measurements
TracerVolume = .04
sheet_name = '20210928_1409'    # Use this for all measurements
```

003_Qcalc.R

Source on Save

```r
require(readxl)
require(lubridate)
require(MESS)

# - reading FL30 raw data
# - transforming data into readable format with signal in mV

# Arguments
path = "~/work/mountain-hydrology-workshop-2026/Fluorometer/KG_G354"

Standard = 50      # g/l - The concentration of the color tracer (Standard solution)
TracerVolume = 0.1  # l - How many liters! were injected into the river?

# ---- 2021 -----
# Input file (Excel)
filename_excel = '2021 - Measurements_G354.xlsx'

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TracerVolume = .025
sheet_name = '20210928_1310'    # Use this for all measurements
TracerVolume = .04
sheet_name = '20210928_1409'    # Use this for all measurements
```

Each line is part of the program and performs **1 task**

Run each line by:

- Put cursor (mouse) into 1st line (or another line)
- Press [Ctrl] + [Enter]

003_Qcalc.R

Source on Save

```r
require(readxl)
require(lubridate)
require(MESS)

# - reading FL30 raw data
# - transforming data into readable format with signal in mV

# Arguments
path = "~/work/mountain-hydrology-workshop-2026/Fluorometer/KG_G354"

Standard = 50        # g/l - The concentration of the color tracer (Standard solution)
TracerVolume = 0.1  # l - How many liters! were injected into the river?

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sheet_name = '20210928_1310'    # Use this for all measurements
TracerVolume = .04
sheet_name = '20210928_1409'    # Use this for all measurements
```

003_Qcalc.R

Source on Save | Run | Source

```r
require(readxl)
require(lubridate)
require(MESS)

# - reading FL30 raw data
# - transforming data into readable format with signal in mV

# Arguments
path = "~/work/mountain-hydrology-workshop-2026/Flu

Standard = 50        # g/l - The concentration of t
TracerVolume = 0.1  # l - How many liters! were inj

# ---- 2021 -----
# Input file (Excel)
filename_excel = '2021 - Measurements_G354.xlsx'

# ----
sheet_name = '20210928_1052'    # Use this for all
TracerVolume = .02
sheet_name = '20210928_1220_cleaned'    # Use this
TracerVolume = .025
sheet_name = '20210928_1310'    # Use this for all
TracerVolume = .04
sheet_name = '20210928_1409'    # Use this for all
TracerVolume = .06
```

2:1 (Top Level)

Console | Terminal | Background Jobs

R 4.3.1 · ~/

```
> require(readxl)
Loading required package: readxl
>
```

Run first line by:

- Put cursor (mouse) into 1st line (or another line)
- Press [Ctrl] + [Enter]

Below ("Console") you see the result.
Here it says:
Loading the package

003_Qcalc.R

Source on Save | Run | Source

```r
require(readxl)
require(lubridate)
require(MESS)

# - reading FL30 raw data
# - transforming data into readable format with signal in mV

# Arguments
path = "~/work/mountain-hydrology-workshop-2026/Fluorometer/KG_G354"

Standard = 50
TracerVolume = 0.1  # l  How many liters! were injected into the rive

# ---- 2021 -----
# Input file (Excel)
filename_excel = '2021 - Measurements_G354.xlsx'

# ----
sheet_name = '20210928_1052'    # Use this for all measurements
TracerVolume = .02
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TracerVolume = .025
sheet_name = '20210928_1310'    # Use this for all measurements
TracerVolume = .04
sheet_name = '20210928_1409'    # Use this for all measurements
TracerVolume = .06
```

2:1 (Top Level)

Console | Terminal | Background Jobs

R 4.3.1 · ~/

```
> require(readxl)
Loading required package: readxl
>
```

The program is designed that you only have to change very few things!

These things are:

- Define the concentration of the base solution (Standard)
- Define how much color was injected (define for each measurement later)
- Define the name of the prepared Fluorometer file (Excel)

003_Qcalc.R

Source on Save | Run | Source

```r
require(readxl)
require(lubridate)
require(MESS)

# - reading FL30 raw data
# - transforming data into readable format with signal in mV

# Arguments
path = "~/work/mountain-hydrology-workshop-2026/Fluorometer/KG_G354"

Standard = 50        # g/l - The concentration of the color tracer (St
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sheet_name = '20210928_1310'    # Use this for all measurements
TracerVolume = .04
sheet_name = '20210928_1409'    # Use this for all measurements
TracerVolume = .06
```

2:1 (Top Level)

Console | Terminal | Background Jobs

R 4.3.1 · ~/

```
> require(readxl)
Loading required package: readxl
>
```

The program is designed that you only have to change very few things!

These things are:

- Define the concentration of the base solution (Standard)
- Define how much color was injected
- Define the name of the prepared Fluorometer file (Excel)
- The name of the spreadsheet (Excel)
- Define how much color was injected

That's it. You don't need to change anything else!

```r
# ------------ NO NEED TO CHANGE ANYTHING HEREAFTER --------------- #
# gc() # This cleans the memory - against JAVA Problems ...
setwd(path)
# source('../R/FUN_install_packages.R')
M = Standard * TracerVolume * 1000 # to mg
# Source all functions for cleaning and preparation
source('../0_R-FUN/FUN_curve-processing-fixedCalibrationLiquid.R')

# ------- READ DATA ------- #
# d_ = read.xlsx(sheetName = sheet_name,  # this one needs to be changed to read in t
#                file = filename_excel,
#                header = F)
library(lubridate)
d_ = read_xlsx_(filename = filename_excel,sheet = sheet_name)
# ---- remove NAs
d_ = rm_nas(d_)
plot(d_$X1, d_$X4, type='l')
require(zoo)
ttt = na.approx(d_$X4)
d_$X4 = ttt
lines(d_$X1, ttt, col='red')
d_ = na.omit(d_)
# ---- Make date-time
tmp_date = make_date_time(d_)

# ---- Combine data
d = data.frame(date=tmp_date, Tracer1=d_$X4)

# ---- Noise removal
d = remove_noise(d)

# ---- Calibration
calibrate(d, 1, 600)

# ---- Relationship mV to ppb
d = relate_mV_ppb(d)


# ---- Discharge Calculation --------- #
# Calculate discharge with the global method
# ppb = 1 µg/L
# Q = M/A
# - M [mg] = 1000 µg
# - A [µg*s / L]
# ->
# L*1000/s = m3/s
# ---------------------------------------- #


# ---- Curve 1
calc_discharge(d)
```

Scroll down to where it says:
"# ------------ NO NEED TO CHANGE ANYTHING HEREAFTER --------------- #

- Click into this line
- Run line by line [Ctrl]+[Enter]
- You will see graphs produced
- You will see messages in the console when you must do something

```r
# ------------ NO NEED TO CHANGE ANYTHING HEREAFTER --------------- #
# gc() # This cleans the memory - against JAVA Problems ...
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M = Standard * TracerVolume * 1000 # to mg
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# ------- READ DATA ------- #
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d_ = read_xlsx_(filename = filename_excel,sheet = sheet_name)
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lines(d_$X1, ttt, col='red')
d_ = na.omit(d_)
# ---- Make date-time
tmp_date = make_date_time(d_)

# ---- Combine data
d = data.frame(date=tmp_date, Tracer1=d_$X4)

# ---- Noise removal
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# ---- Calibration
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# - A [µg*s / L]
# ->
# L*1000/s = m3/s
# ---------------------------------------- #


# ---- Curve 1
calc_discharge(d)
```

Shows the measurement

Action: “Noise removal”

Action: “Calibration”

Action: “Calculate Discharge”

Scroll down to where it says:
“# ------------ NO NEED TO CHANGE ANYTHING HEREAFTER --------------- #

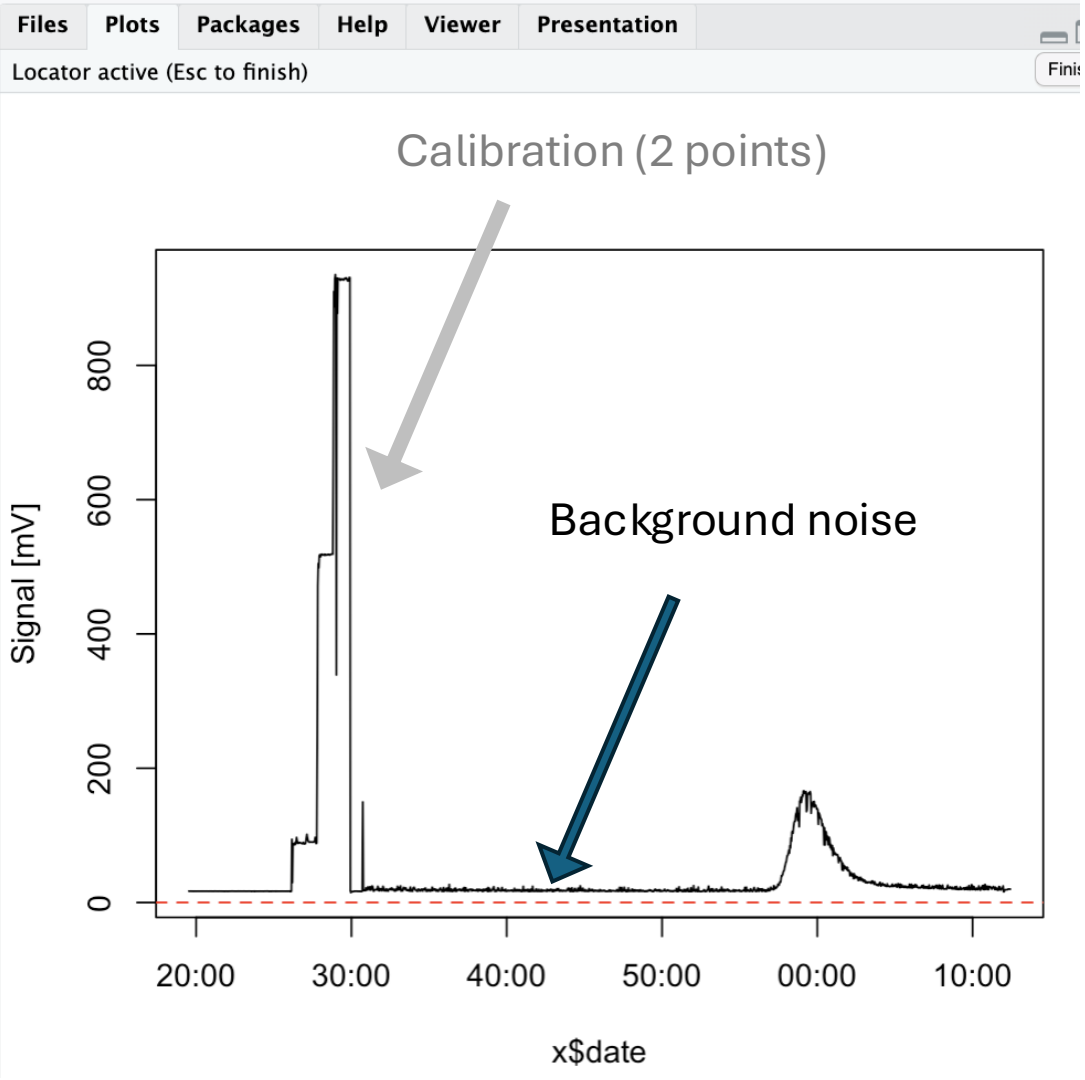
- Click into this line
- Run line by line [Ctrl]+[Enter]
- You will see graphs produced
- You will see messages in the console when you must do something

The 3 actions are explained on the next slides

Action: “Noise removal”

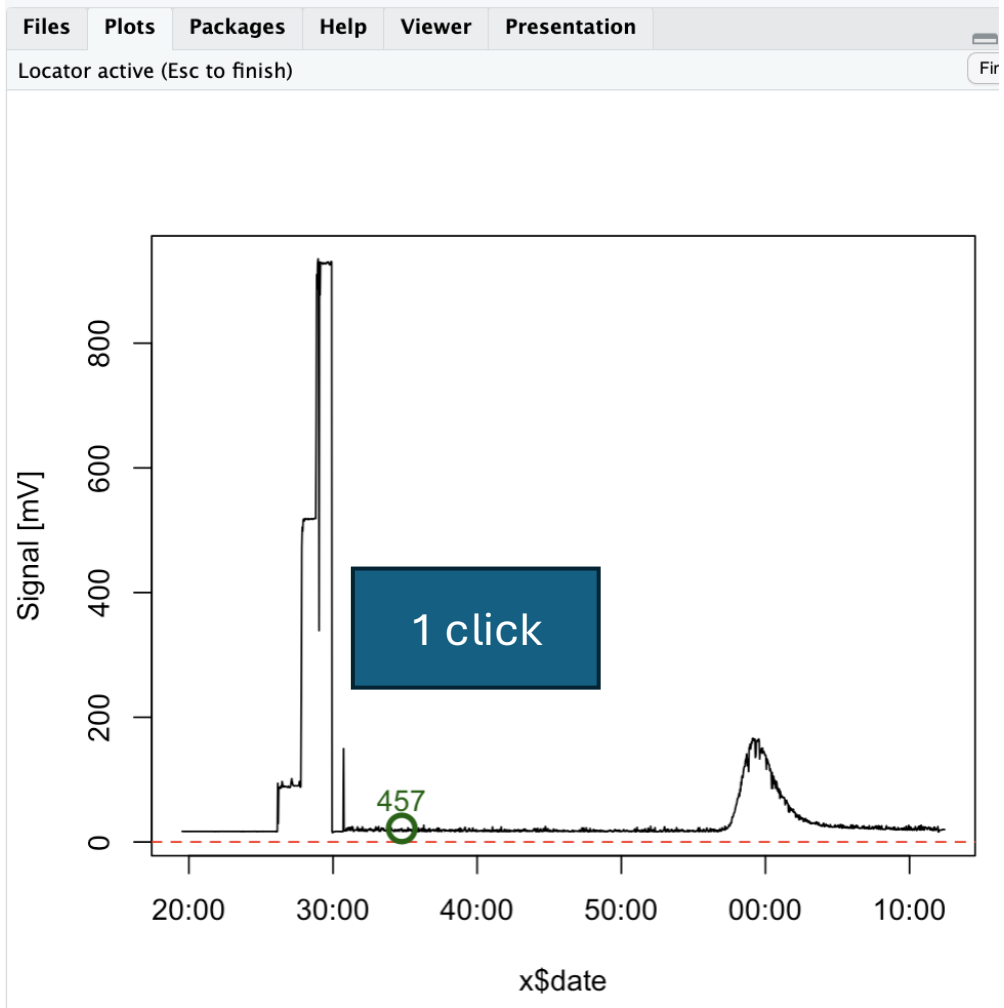
```
> d = remove_noise(d)
[1] "Select first point (1/2) in the plot (click) to identify background noise ... Press [Esc] to interrupt!"
```

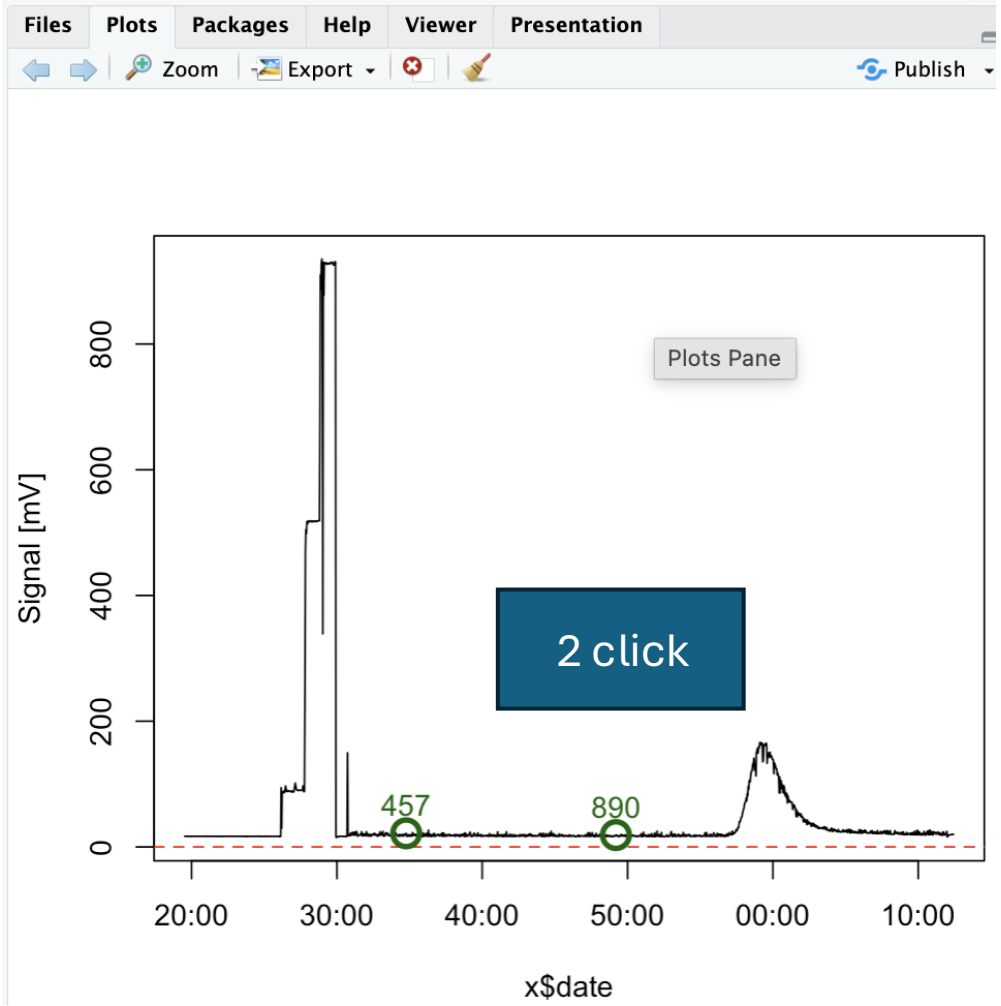
Click into the graph to select 2 points to identify the background noise (electrical signal that is always present)



Action: “Noise removal”

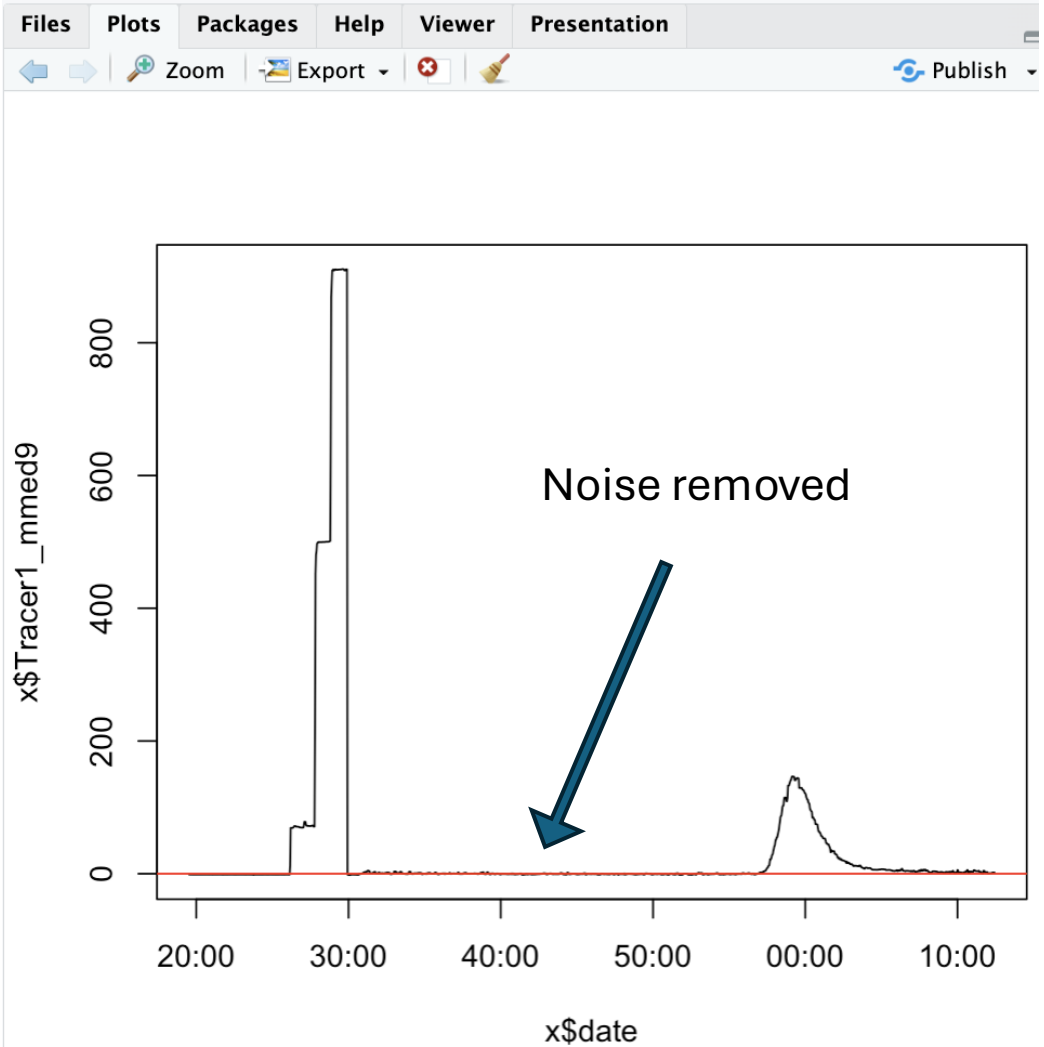
```
> d = remove_noise(d)
[1] "Select first point (1/2) in the plot (click) to identify background noise ... Press [Esc] to interrupt!"
```





Action: “Noise removal”

```
> # ---- Noise removal
> d = remove_noise(d)
[1] "Select first point (1/2) in the plot (click) to identify background noise ... Press [Esc] to interrupt!"
[1] "Select second point (2/2) in the plot (click) to identify background noise ... Press [Esc] to interrupt!"
[1] "Background noise signal is: 18.18 mV"
[1] "Background noise removed."
>
```

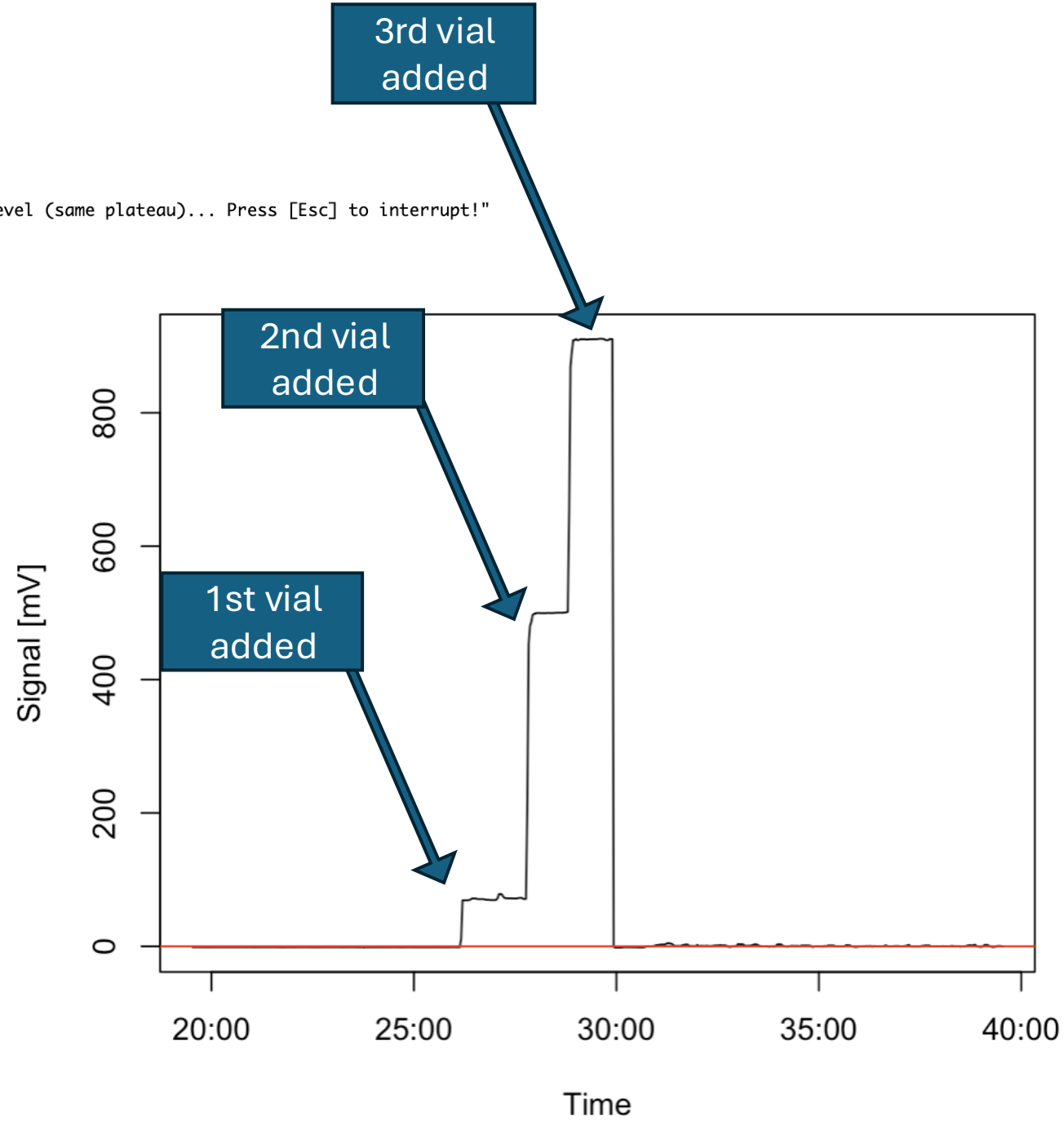


Action: “Calibration”

```
> # ---- Calibration
> calibrate(d, 1, 600)
[1] "---- Identifying calibration levels ----"
[1] "Select first point (1/2) in the plot (click) to identify first calibration level (same plateau)... Press [Esc] to interrupt!"
```

3 calibration points?!
Check your notebook!

(here was a contamination in the bucket – we fix this later. For now we pretend that we had 3 points)



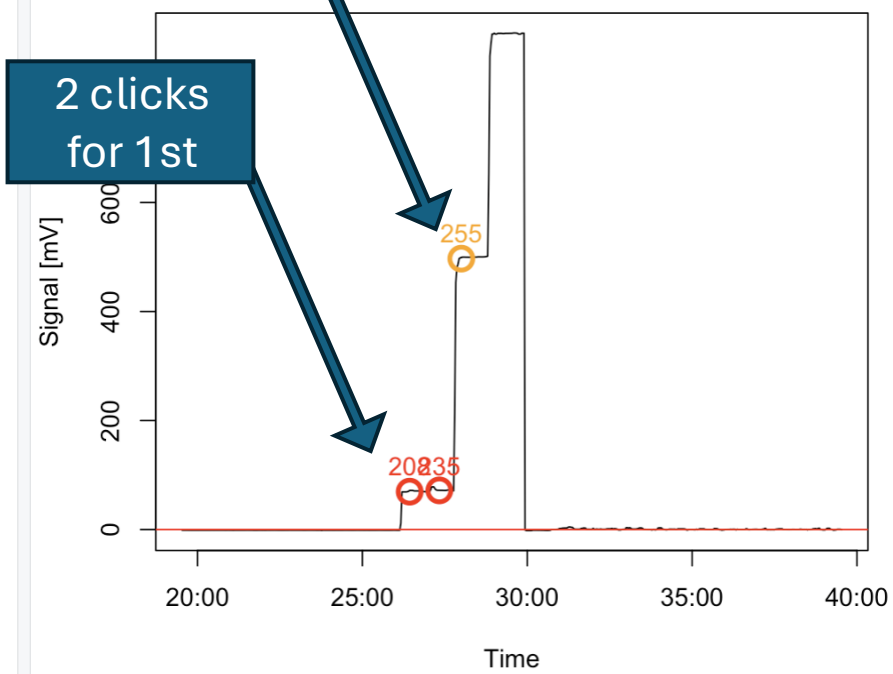
Action: “Calibration”

```
> require(zoo)
> ttt = na.approx(d_$X4)
> d_$X4 = ttt
> lines(d_$X1, ttt, col='red')
> d_ = na.omit(d_)
> # ---- Make date-time
> tmp_date = make_date_time(d_)
[1] "Time series looks fine."
> # ---- Combine data
> d = data.frame(date=tmp_date, Tracer1=d_$X4)
> # ---- Noise removal
> d = remove_noise(d)
nd noise ... Press [Esc] to interrupt!"
und nois ... Press [Esc] to interrupt!"
> calibrate(d, 1, 600)
[1] "---- Identifying calibration levels ----"
[1] "Select first point (1/2) in the plot (click) to identify first calibration level (same plateau)... Press [Esc] to interrupt!"
[1] "Select second point (2/2) in the plot (click) to identify first calibration level (same plateau)... Press [Esc] to interrupt!"
[1] "Calibration signal (first level) is: 71.77 mV"
[1] "Select first point (1/2) in the plot (click) to identify second calibration level (same plateau)... Press [Esc] to interrupt!"
[1] "Select second point (2/2) in the plot (click) to identify second calibration level (same plateau)... Press [Esc] to interrupt!"
```

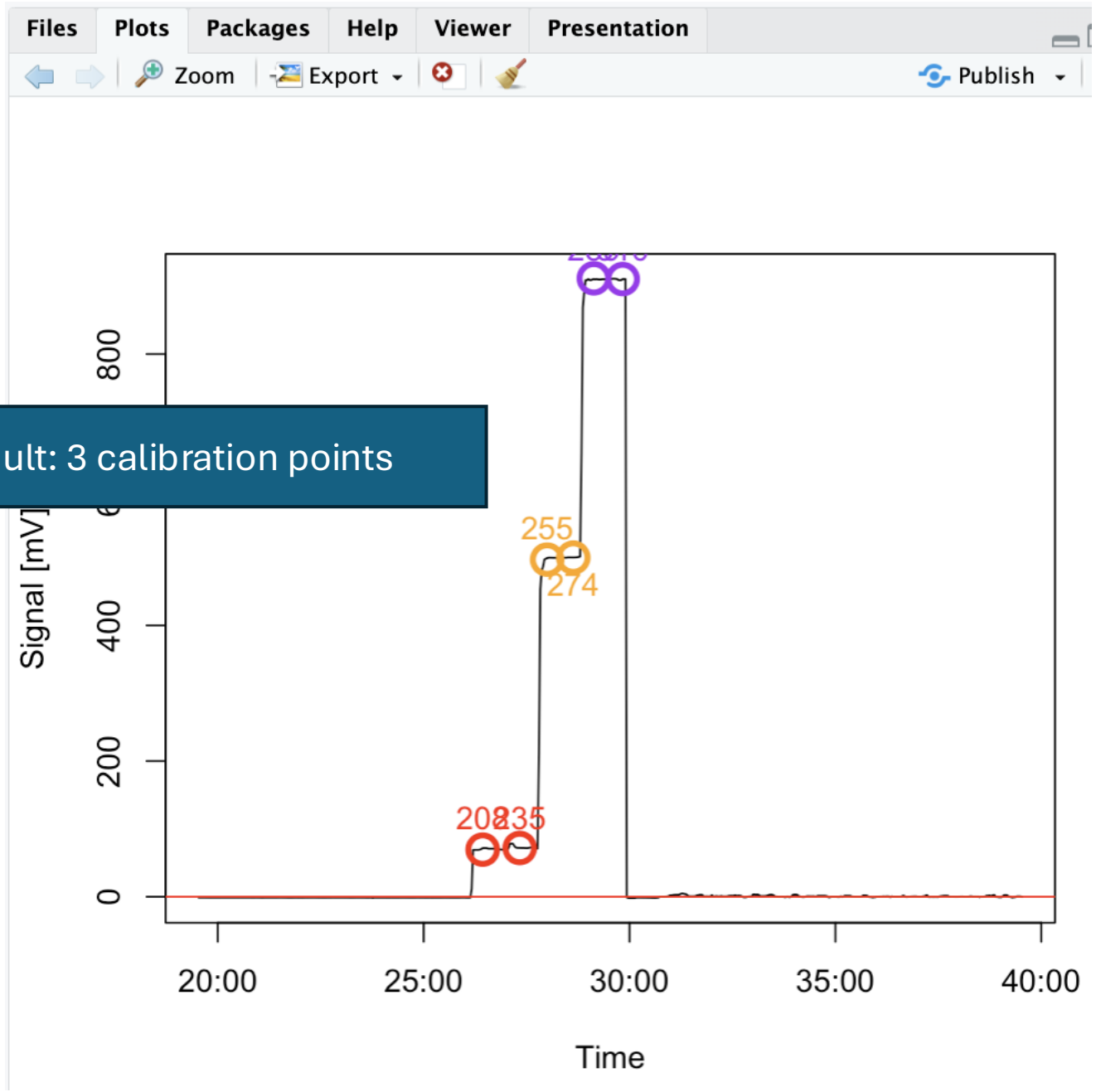
Click to always identify start and end of each calibration point

2 clicks for 2nd

2 clicks for 1st



Action: “Calibration”

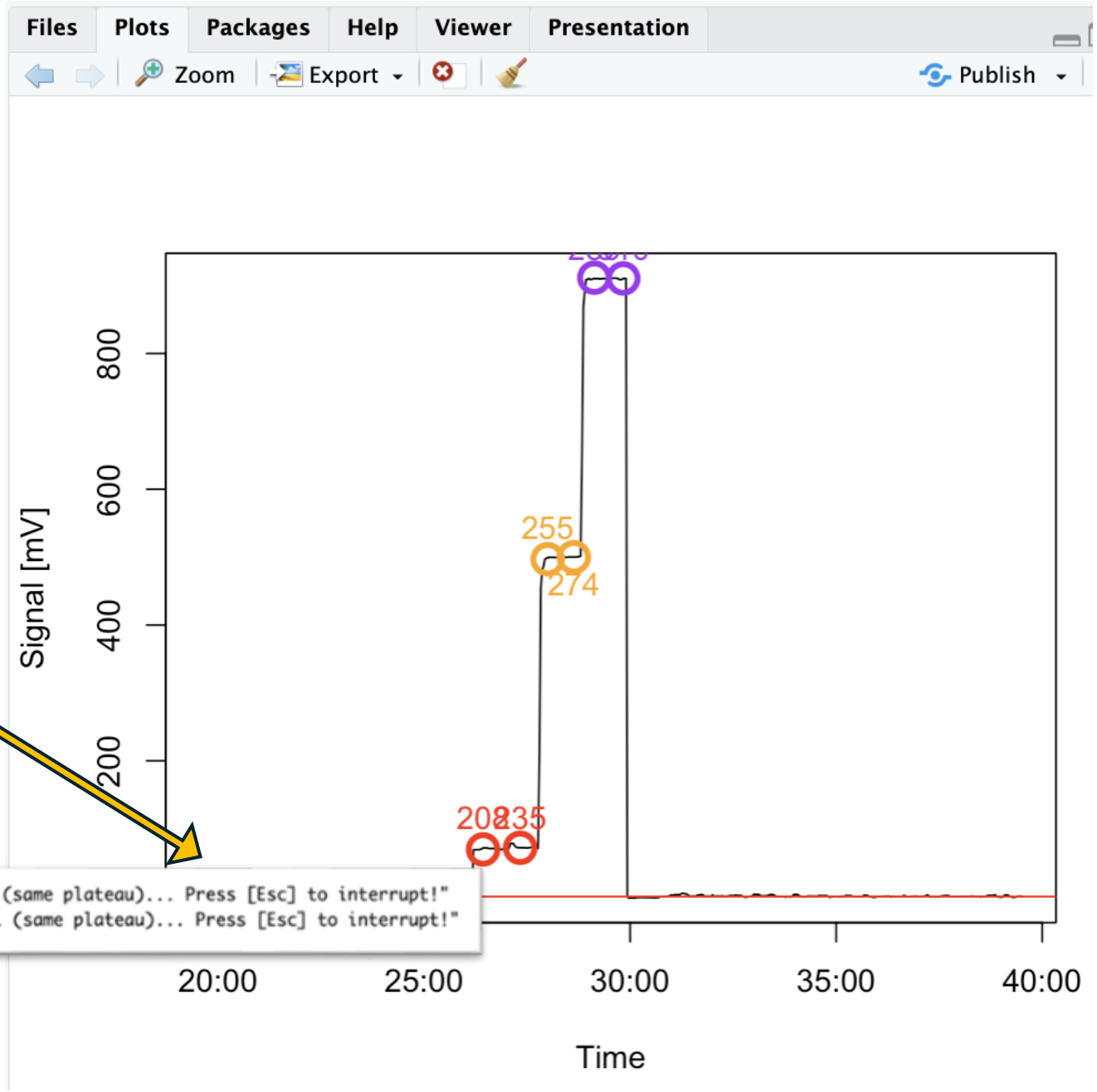


Default: 3 calibration points

Action: “Calibration”

Have only 2 calibration points?

You can always press [Esc] to stop! For example, if you have only 2 or even 1 point.




```
[1] "Select first point (1/2) in the plot (click) to identify second calibration level (same plateau)... Press [Esc] to interrupt!"
[1] "Select second point (2/2) in the plot (click) to identify second calibration level (same plateau)... Press [Esc] to interrupt!"
```

Action: “Calibration”

Check the results

```
# ---- Relationship mV to ppb
d = relate_mV_ppb(d)
```

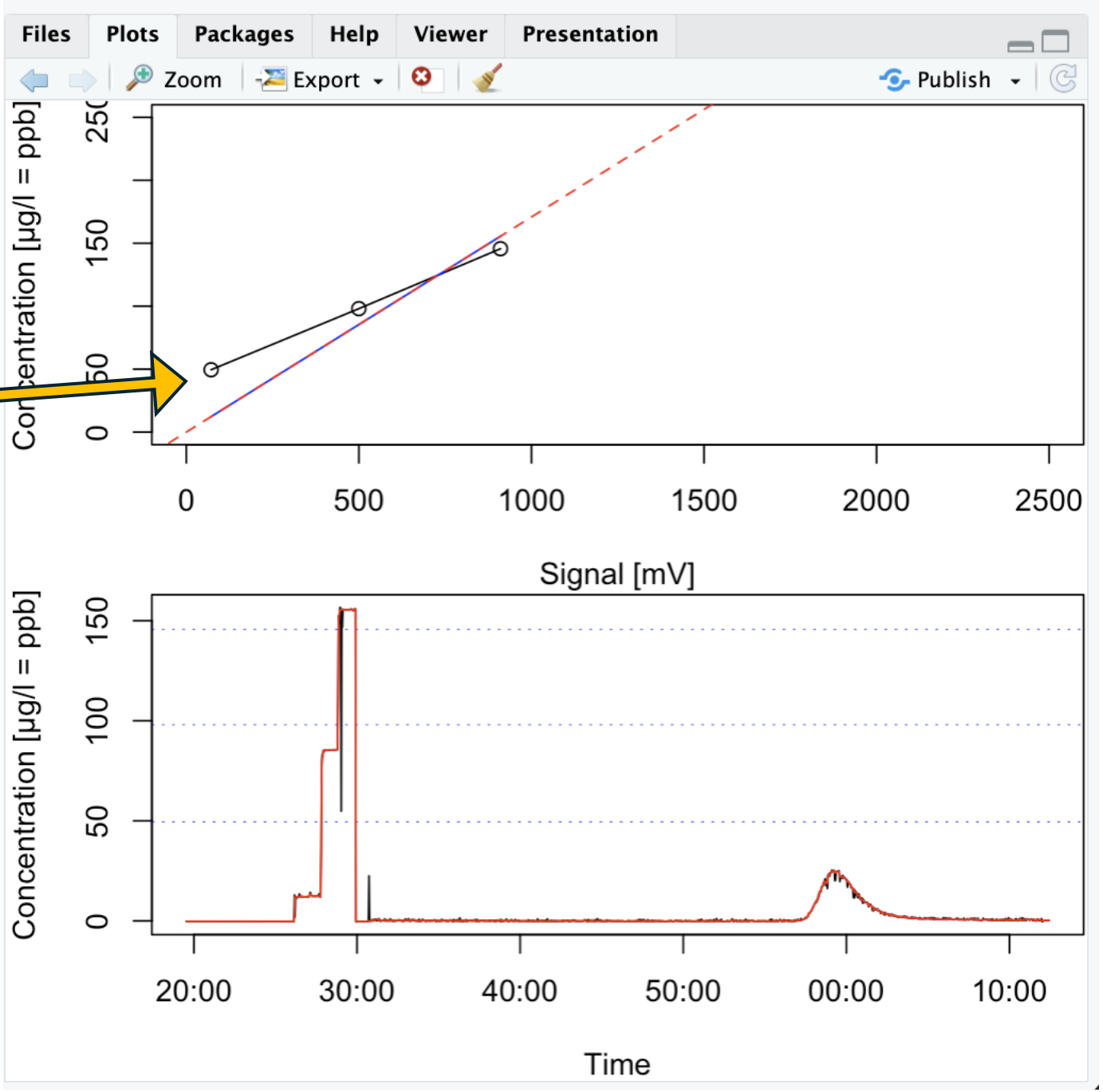
- Click into this line
- Run [Ctrl]+[Enter]

Action: “Calibration”

Check the results

Are the points on the red line?

```
# ---- Relationship mV to ppb
d = relate_mV_ppb(d)
```

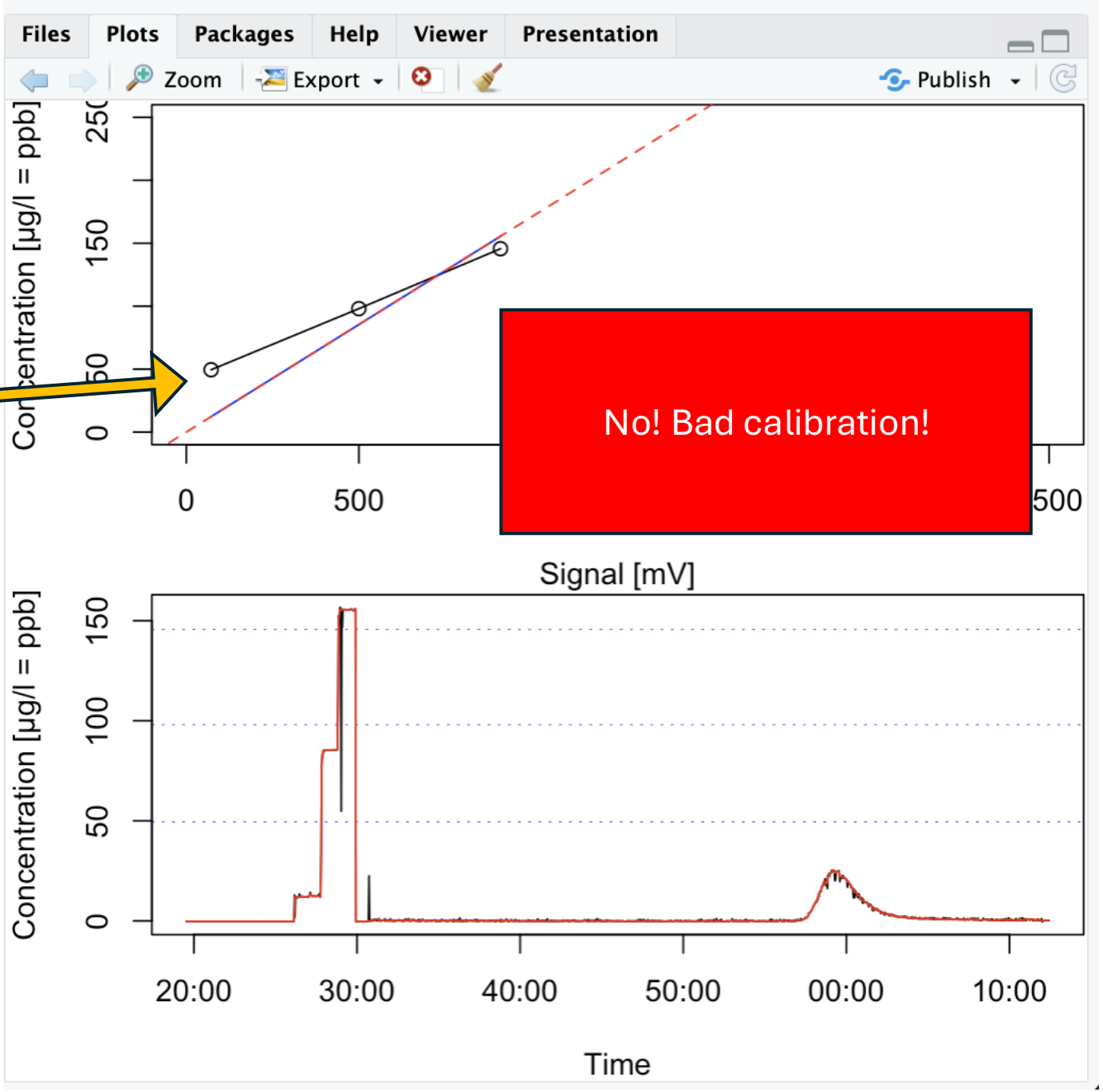


Action: “Calibration”

Check the results

Are the points on the red line?

```
# ---- Relationship mV to ppb
d = relate_mV_ppb(d)
```



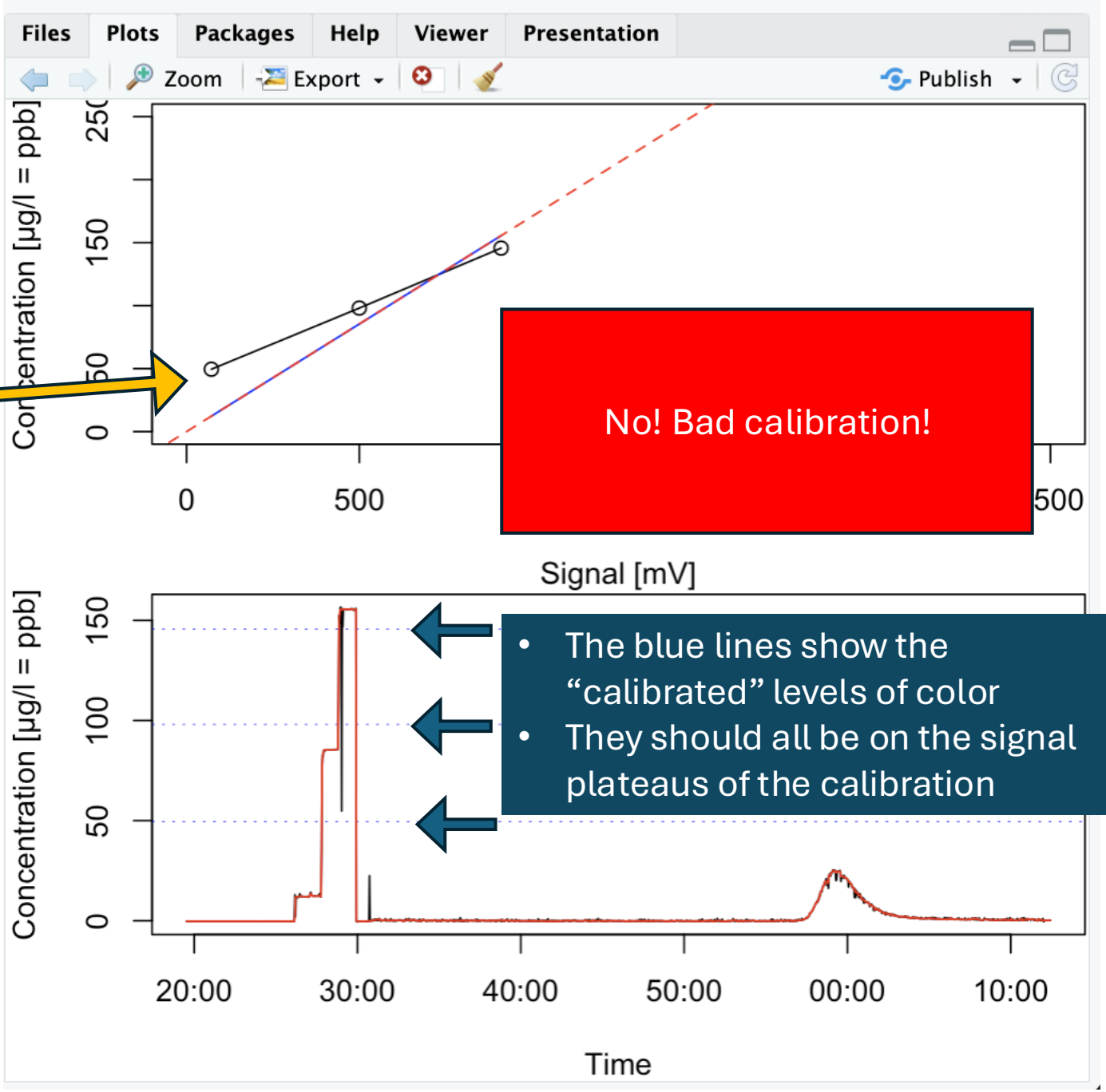
No! Bad calibration!

Action: “Calibration”

Check the results

Are the points on the red line?

```
# ---- Relationship mV to ppb
d = relate_mV_ppb(d)
```



No! Bad calibration!

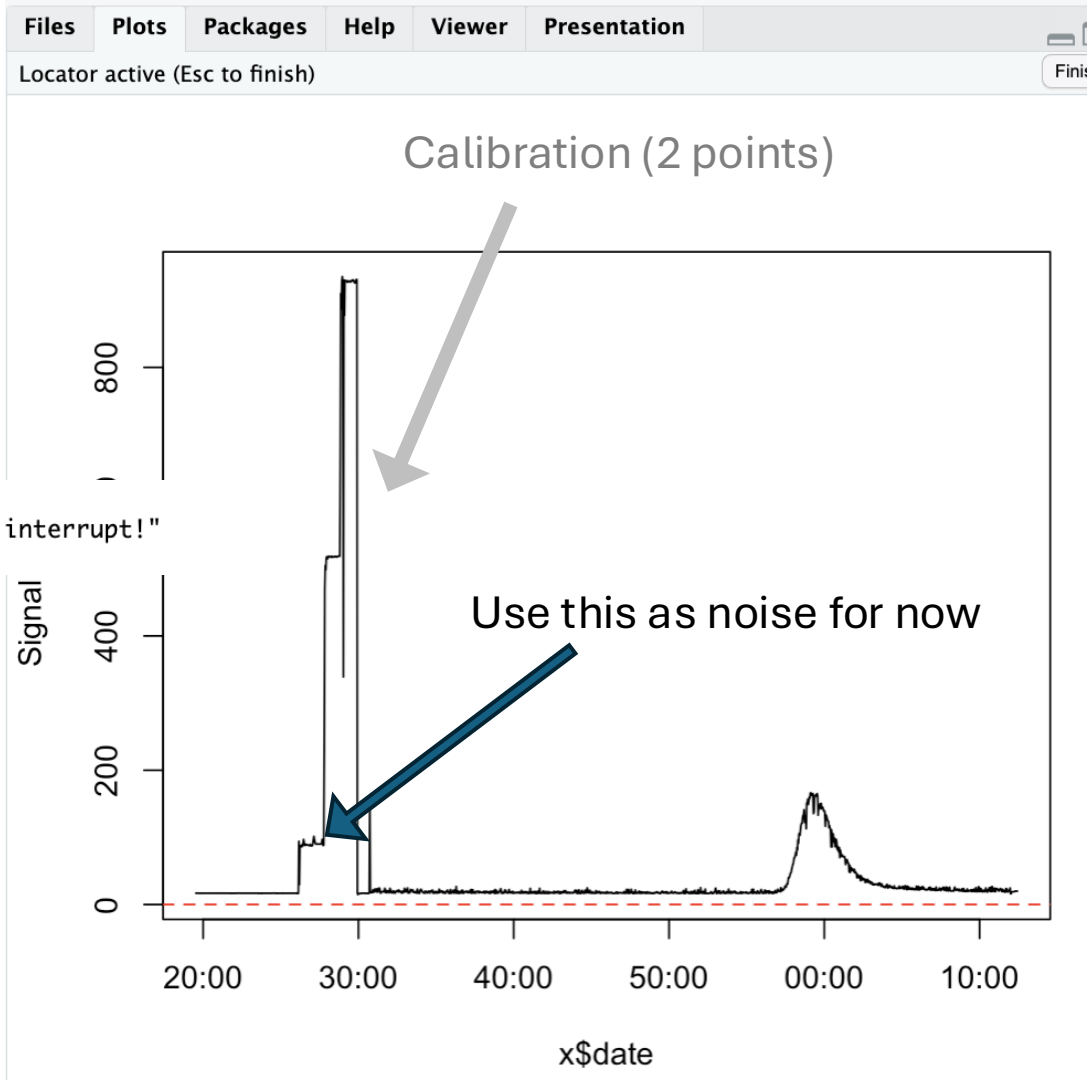
- The blue lines show the “calibrated” levels of color
- They should all be on the signal plateaus of the calibration

Action: “Calibration”

Fixing calibration problems:

- Go back to noise removal
- Run line again [Ctrl] + [Enter]
- Identify the contamination as noise

```
> d = remove_noise(d)
[1] "Select first point (1/2) in the plot (click) to identify background noise ... Press [Esc] to interrupt!"
```

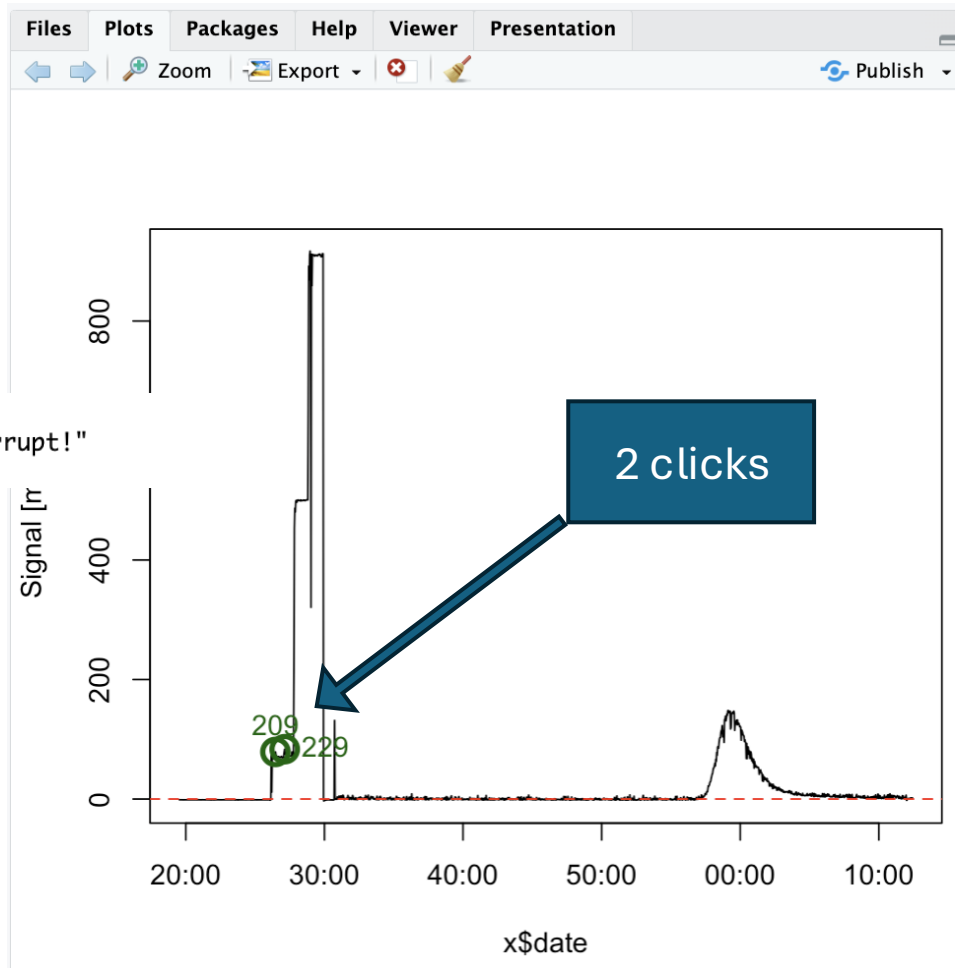


Action: “Calibration”

Fixing calibration problems:

- Go back to noise removal
- Run line again [Ctrl] + [Enter]
- Identify the contamination as noise

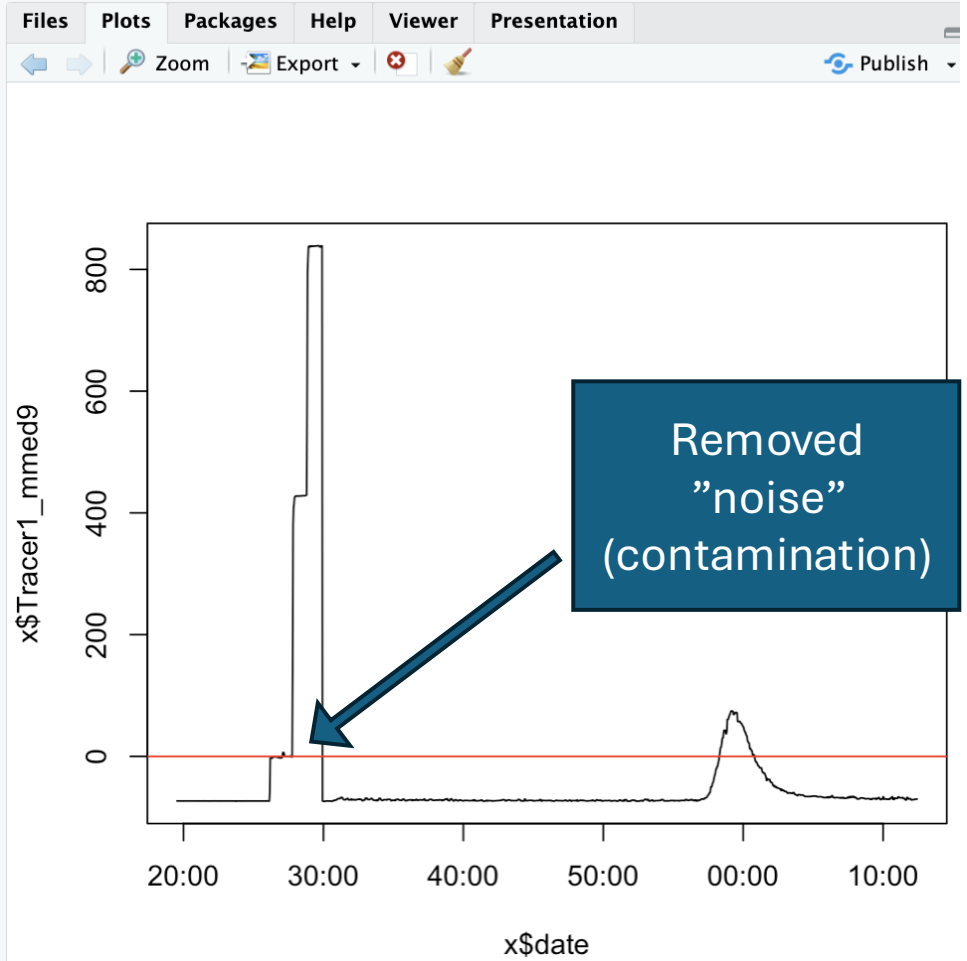
```
> d = remove_noise(d)
[1] "Select first point (1/2) in the plot (click) to identify background noise ... Press [Esc] to interrupt!"
```



Action: “Calibration”

Fixing calibration problems:

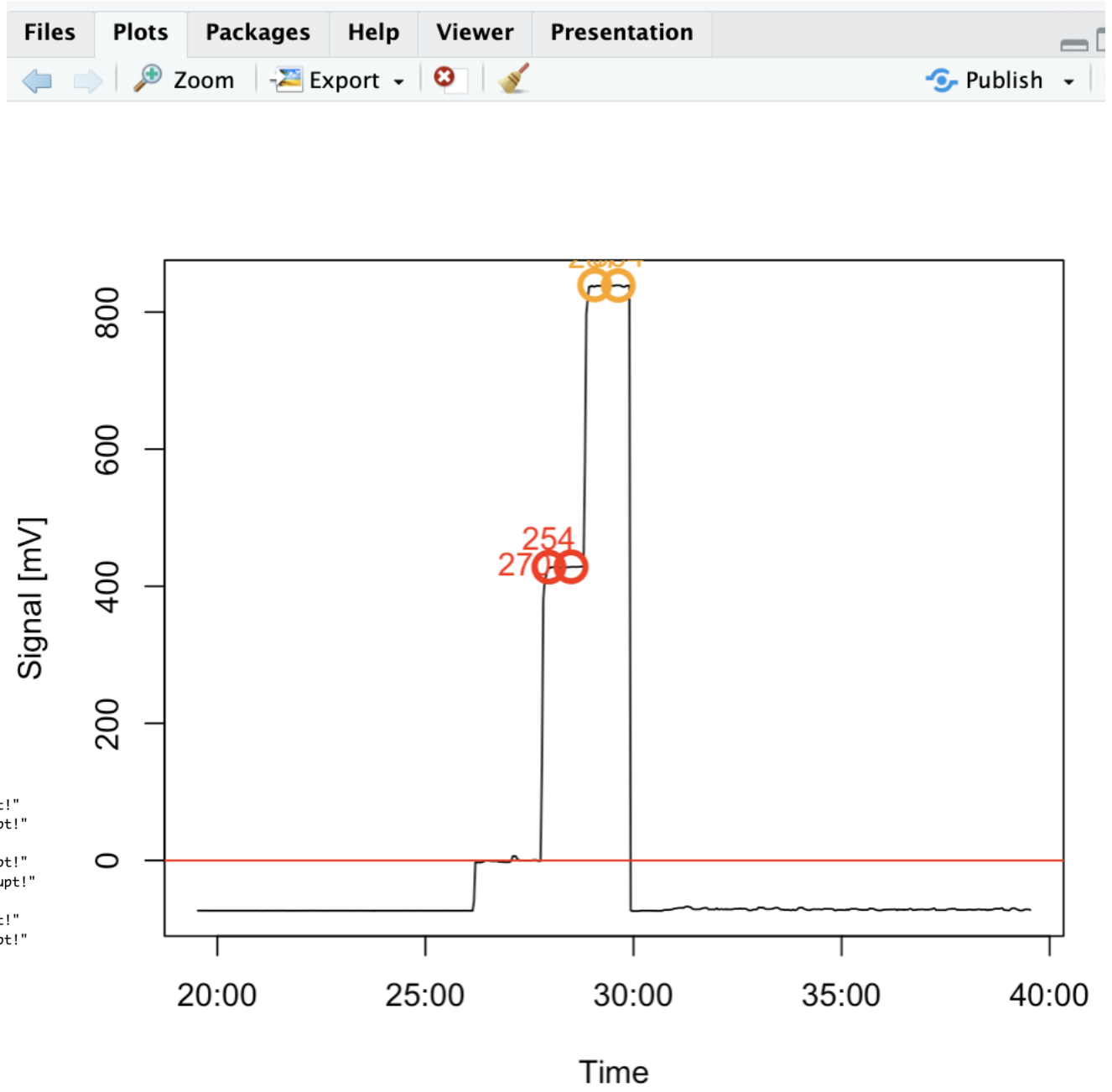
- Go back to noise removal
- Run line again [Ctrl] + [Enter]
- Identify the contamination as noise



Action: “Calibration”

Fixing calibration problems:

- Re-run the calibration (2 points)
- Interrupt the 3rd point (press [Esc] twice)




```
> # ---- Calibration
> calibrate(d, 1, 600)
[1] "---- Identifying calibration levels ----"
[1] "Select first point (1/2) in the plot (click) to identify first calibration level (same plateau)... Press [Esc] to interrupt!"
[1] "Select second point (2/2) in the plot (click) to identify first calibration level (same plateau)... Press [Esc] to interrupt!"
[1] "Calibration signal (first level) is: 427.69 mV"
[1] "Select first point (1/2) in the plot (click) to identify second calibration level (same plateau)... Press [Esc] to interrupt!"
[1] "Select second point (2/2) in the plot (click) to identify second calibration level (same plateau)... Press [Esc] to interrupt!"
[1] "Calibration signal (second level) is: 838.33 mV"
[1] "Select first point (1/2) in the plot (click) to identify third calibration level (same plateau)... Press [Esc] to interrupt!"
[1] "Select second point (2/2) in the plot (click) to identify third calibration level (same plateau)... Press [Esc] to interrupt!"
[1] "No third calibration level determined."
[1] "Number of calibration levels: 2"
[1] "Using two calibration points for concetrations of 49.5 & 98.04 ppb and signals of 427.69 & 838.33 mV"
>
```

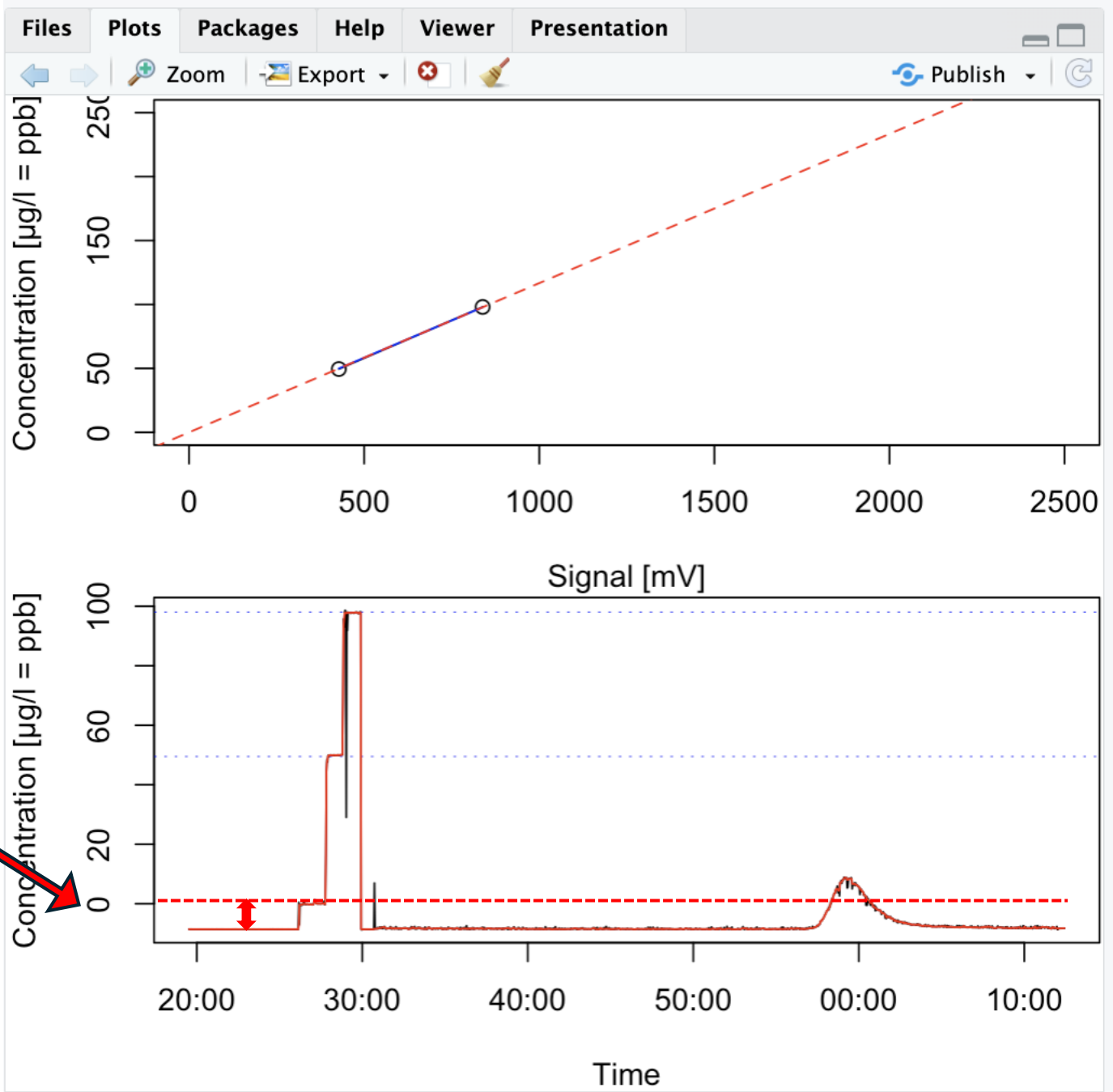
Action: “Calibration”

Fixing calibration problems:

Yes! Good calibration!

(But the background of the normal signal is now negative!)

```
# ---- Relationship mV to ppb
d = relate_mV_ppb(d)
```

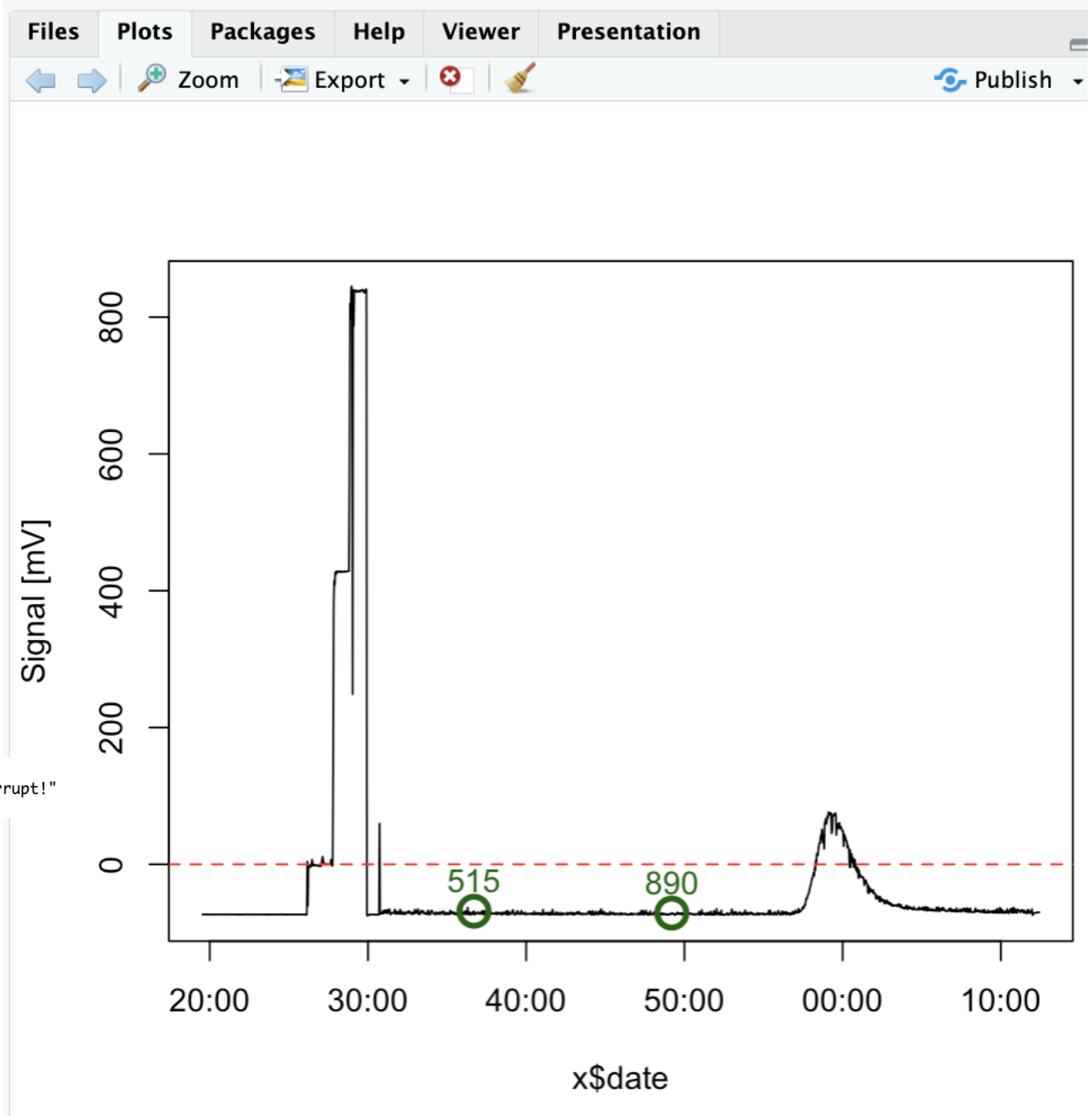


Action: “Calibration”

Fixing calibration problems:

→ Re-run (last time) noise removal again for the main signal!

```
> d = remove_noise(d)
[1] "Select first point (1/2) in the plot (click) to identify background noise ... Press [Esc] to interrupt!"
```

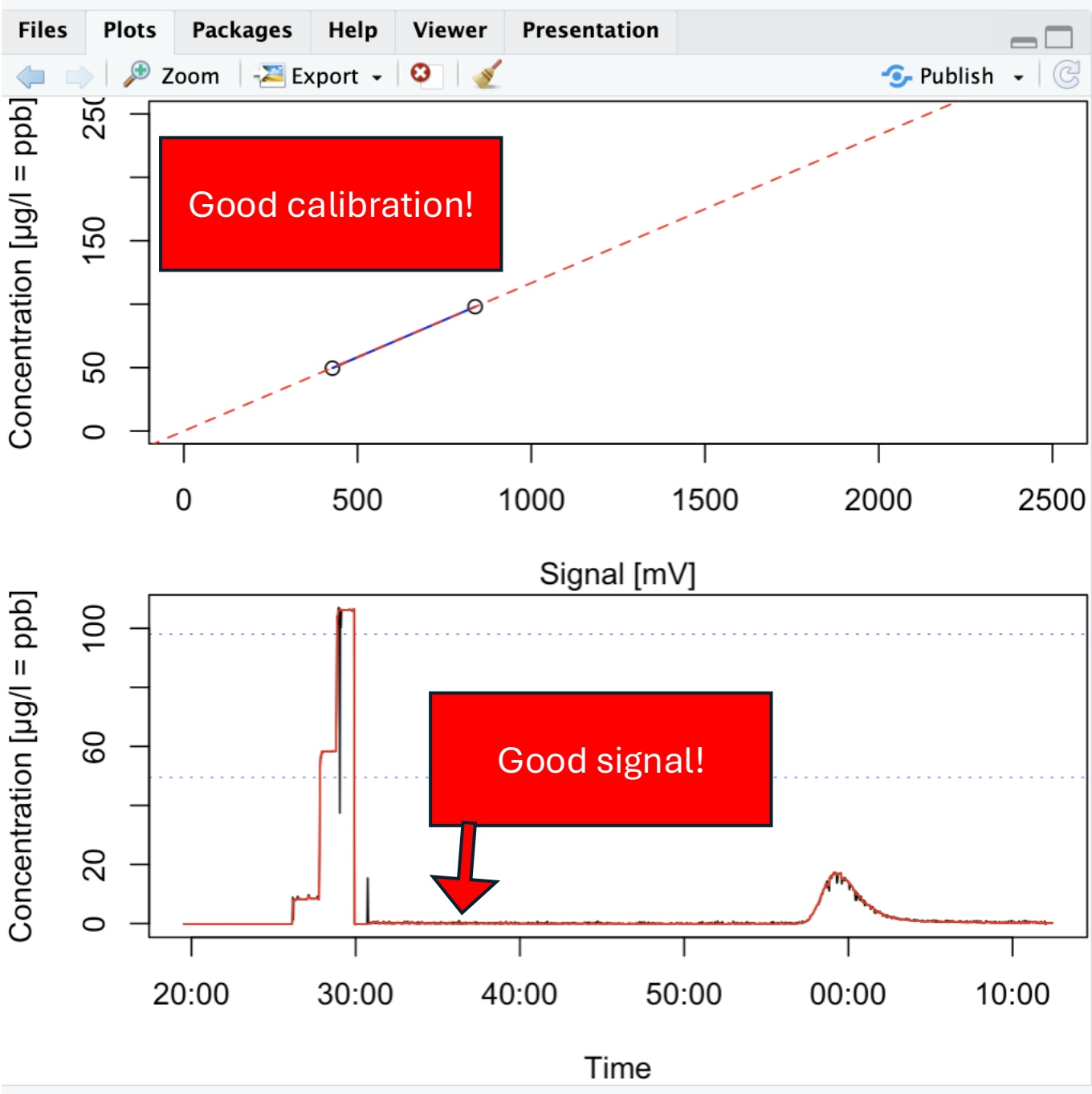


Action: “Calibration”

Fixing calibration problems:

→ Re-run (last time) noise removal again for the main signal!

```
# ---- Relationship mV to ppb
d = relate_mV_ppb(d)
```



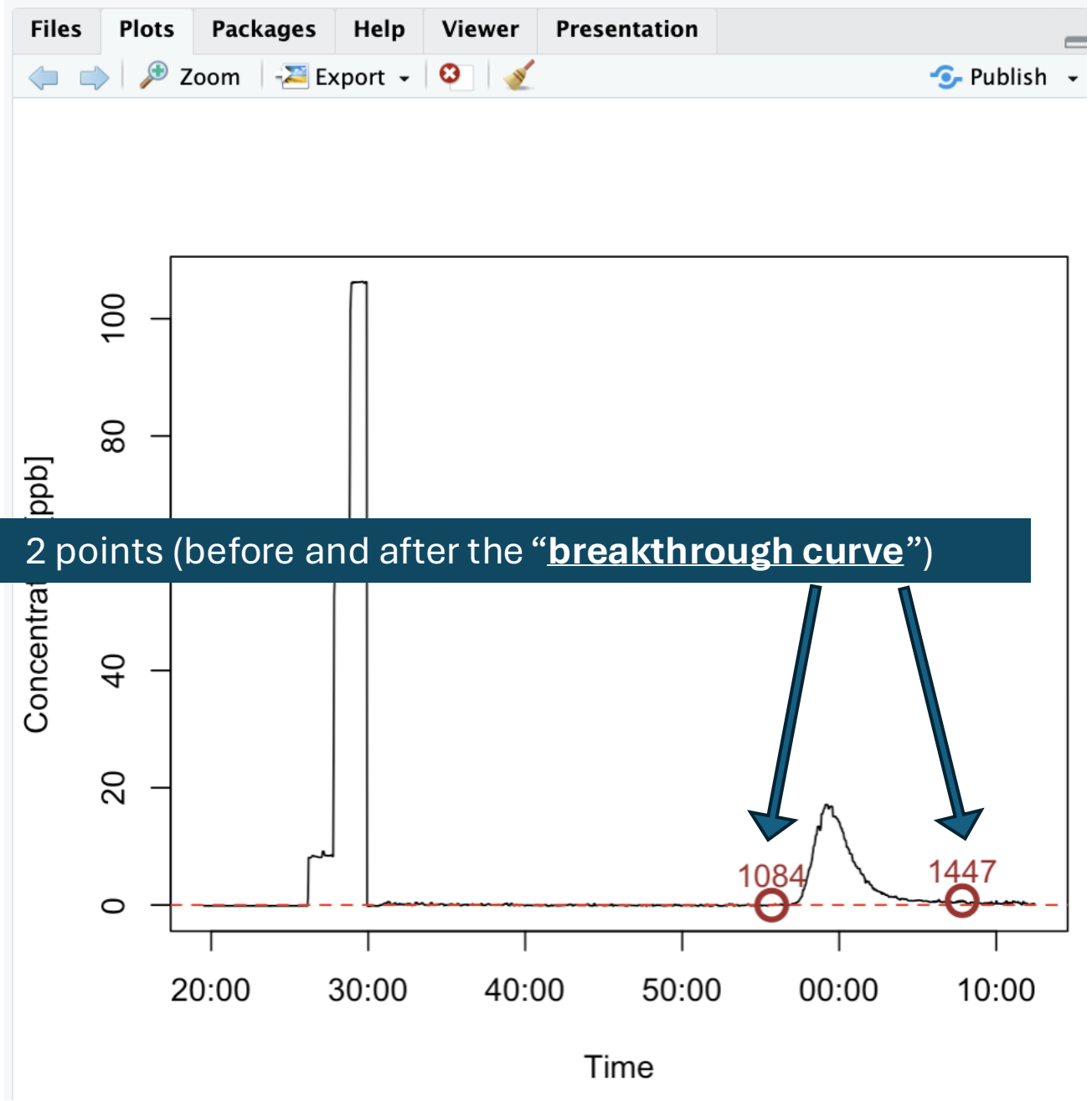
Action: “Calculate Discharge”

```
# ---- Discharge Calculation --------- #
# Calculate discharge with the global method
# ppb = 1 µg/L
# Q = M/A
# - M [mg] = 1000 µg
# - A [µg*s / L]
# ->
# L*1000/s = m3/s
# ---------------------------------------- #


# ---- Curve 1
calc_discharge(d)
```

- Click into this line
- Run [Ctrl]+[Enter]

Action: “Calculate Discharge”

```r
# ---- Discharge Calculation --------- #
# Calculate discharge with the global method
# ppb = 1 µg/L
# Q = M/A
# - M [mg] = 1000 µg
# - A [µg*s / L]
# ->
# L*1000/s = m3/s
# ------------------------------------- #


# ---- Curve 1
calc_discharge(d)
```

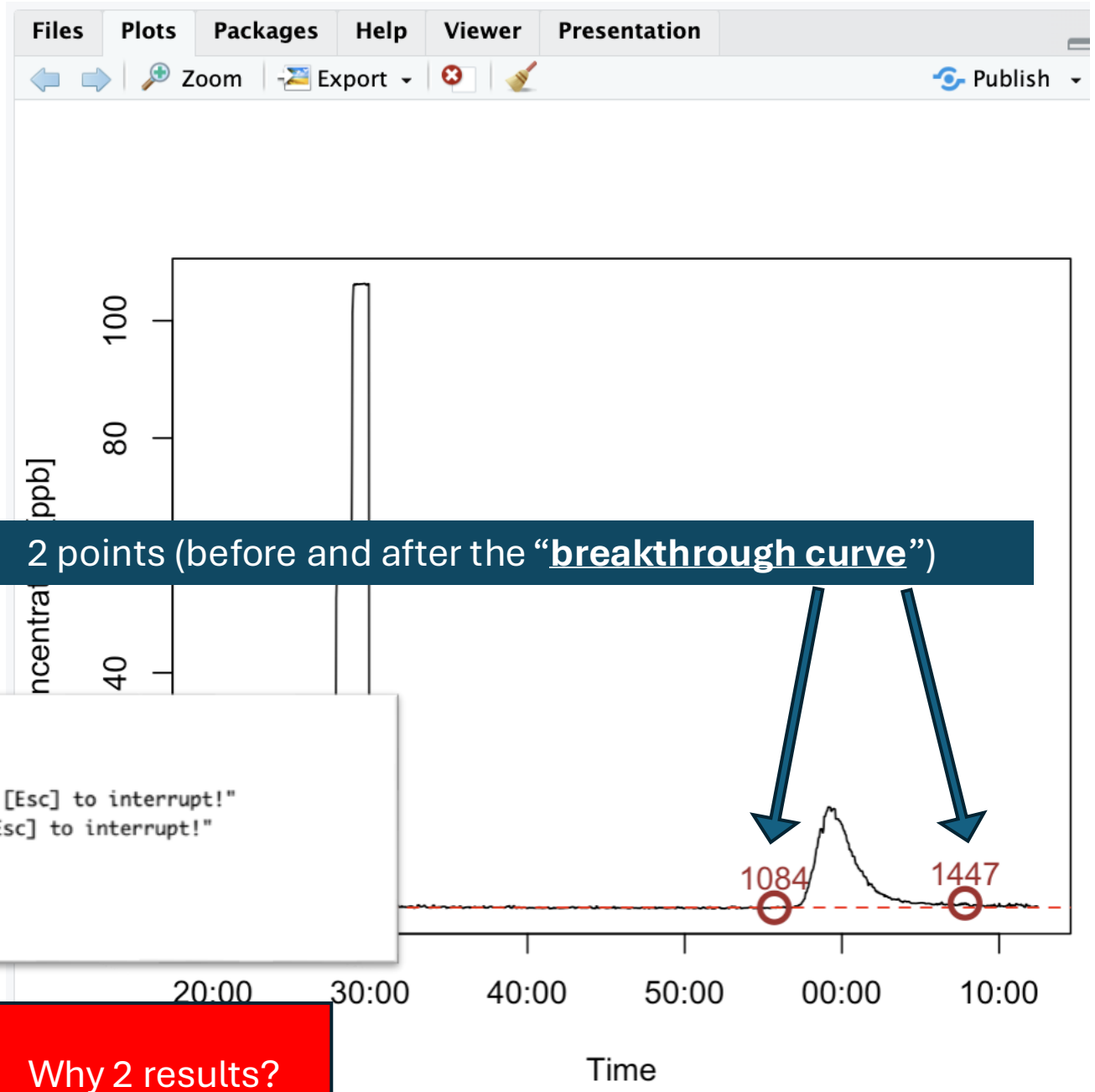


- 2 points (before and after the “**breakthrough curve**”)

Action: “Calculate Discharge”

```r
# ---- Discharge Calculation --------- #
# Calculate discharge with the global method
# ppb = 1 µg/L
# Q = M/A
# - M [mg] = 1000 µg
# - A [µg*s / L]
# ->
# L*1000/s = m3/s
# ------------------------------------- #


# ---- Curve 1
calc_discharge(d)
```

```
> # ---- Curve 1
There were 12 warnings (use warnings() to see them)
> calc_discharge(d)
[1] "Select start point (1/2) in the plot (click) of breakthrough curve ... Press [Esc] to interrupt!"
[1] "Select end point (2/2) in the plot (click) of breakthrough curve ... Press [Esc] to interrupt!"
[1] "Parameters: Tracer mass=6000 mg  ; Standard=50 g/l  ;  Tracer volume=0.02 l"
[1] "Calculated discharge is: 2.00116 m^3/s"
[1] "Calculated discharge (with extrapolation) is: 1.83111 m^3/s"
>
```



- 2 points (before and after the “**breakthrough curve**”)

The result(s)

Why 2 results?

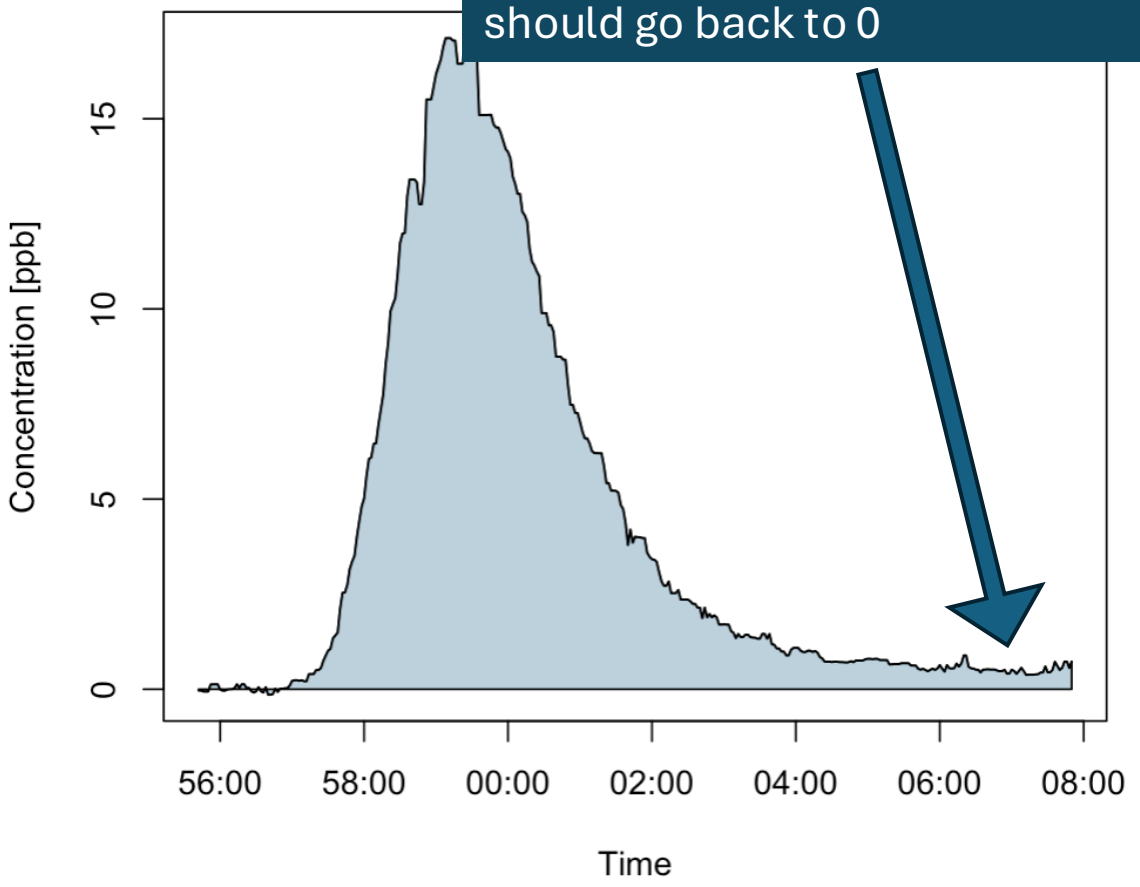
Action: “Calculate Discharge”

```
> # ---- Curve 1
There were 12 warnings (use warnings() to see them)
> calc_discharge(d)
[1] "Select start point (1/2) in the plot (click) of breakthrough curve ... Press [Esc] to interrupt!"
[1] "Select end point (2/2) in the plot (click) of breakthrough curve ... Press [Esc] to interrupt!"
[1] "Parameters: Tracer mass=6000 mg  ; Standard=50 g/l  ;  Tracer volume=0.02 l"
[1] "Calculated discharge is: 2.00116 m^3/s"
[1] "Calculated discharge (with extrapolation) is: 1.83111 m^3/s"
>
```

ges Help Viewer Presentation

Export Publish

The “**breakthrough curve**” should go back to 0



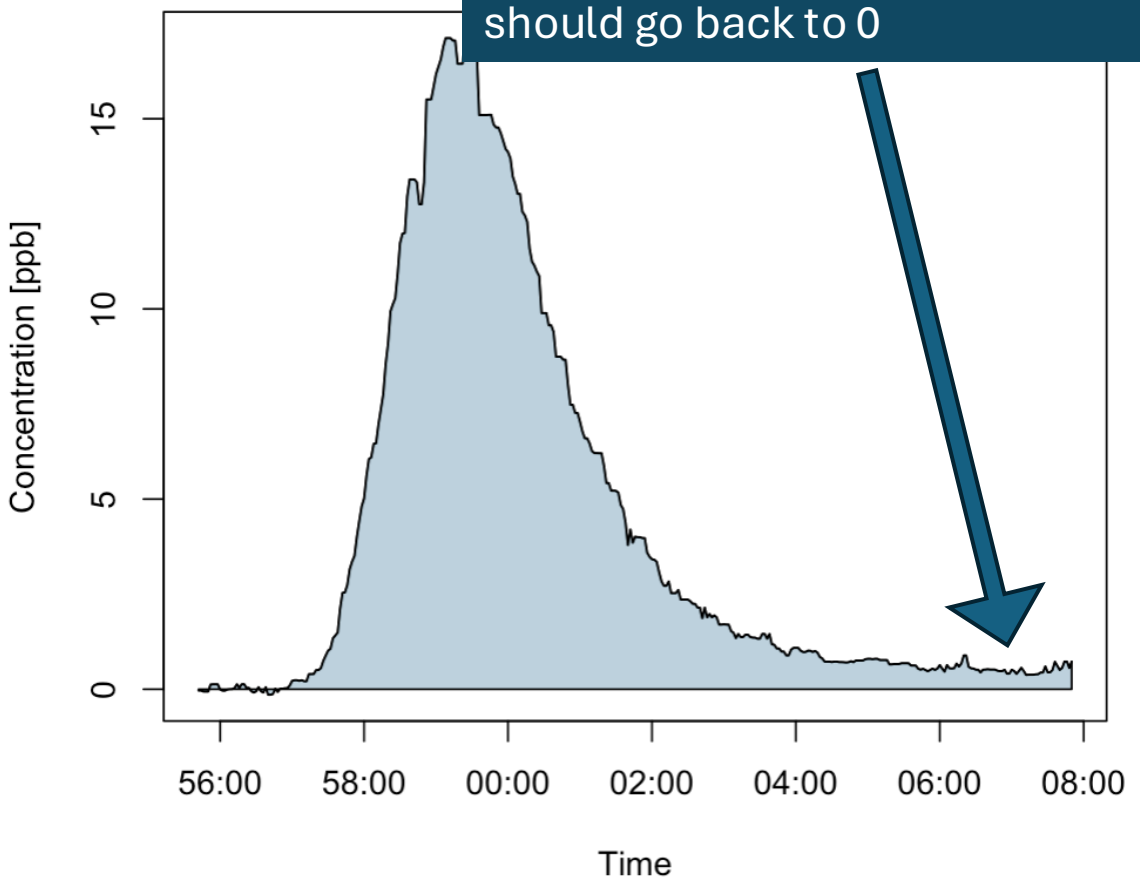
Action: “Calculate Discharge”

```
> # ---- Curve 1
There were 12 warnings (use warnings() to see them)
> calc_discharge(d)
[1] "Select start point (1/2) in the plot (click) of breakthrough curve ... Press [Esc] to interrupt!"
[1] "Select end point (2/2) in the plot (click) of breakthrough curve ... Press [Esc] to interrupt!"
[1] "Parameters: Tracer mass=6000 mg  ; Standard=50 g/l  ;  Tracer volume=0.02 l"
[1] "Calculated discharge is: 2.00116 m^3/s"
[1] "Calculated discharge (with extrapolation) is: 1.83111 m^3/s"
>
```

ges Help Viewer Presentation

Export Publish

The “**breakthrough curve**” should go back to 0



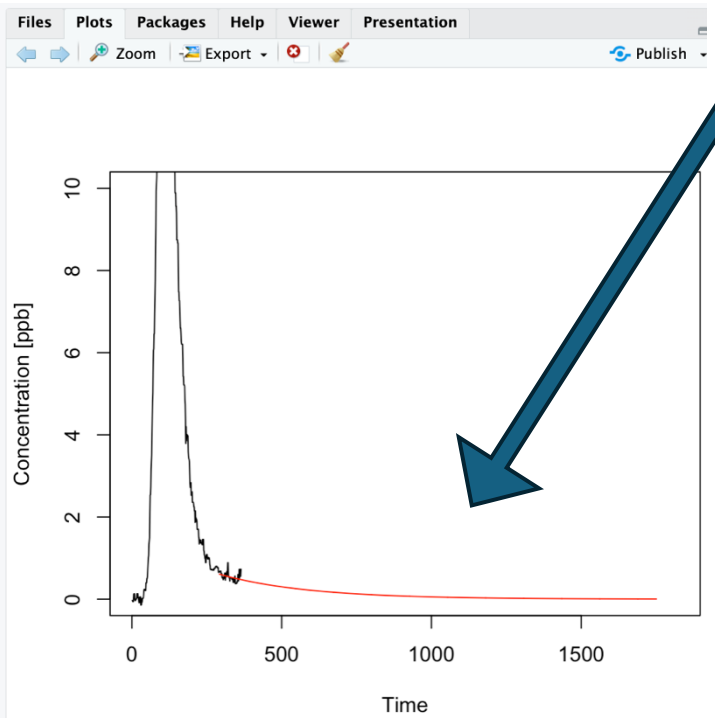
If the signal is in the end still higher than the background (0) → Still color in the river!

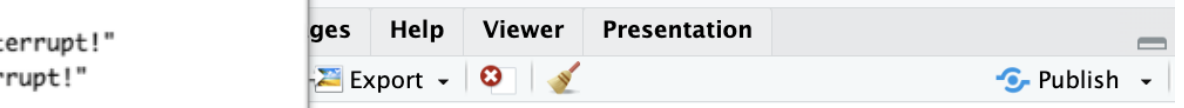
Lower color concentration → higher calculated discharge

Action: “Calculate Discharge”

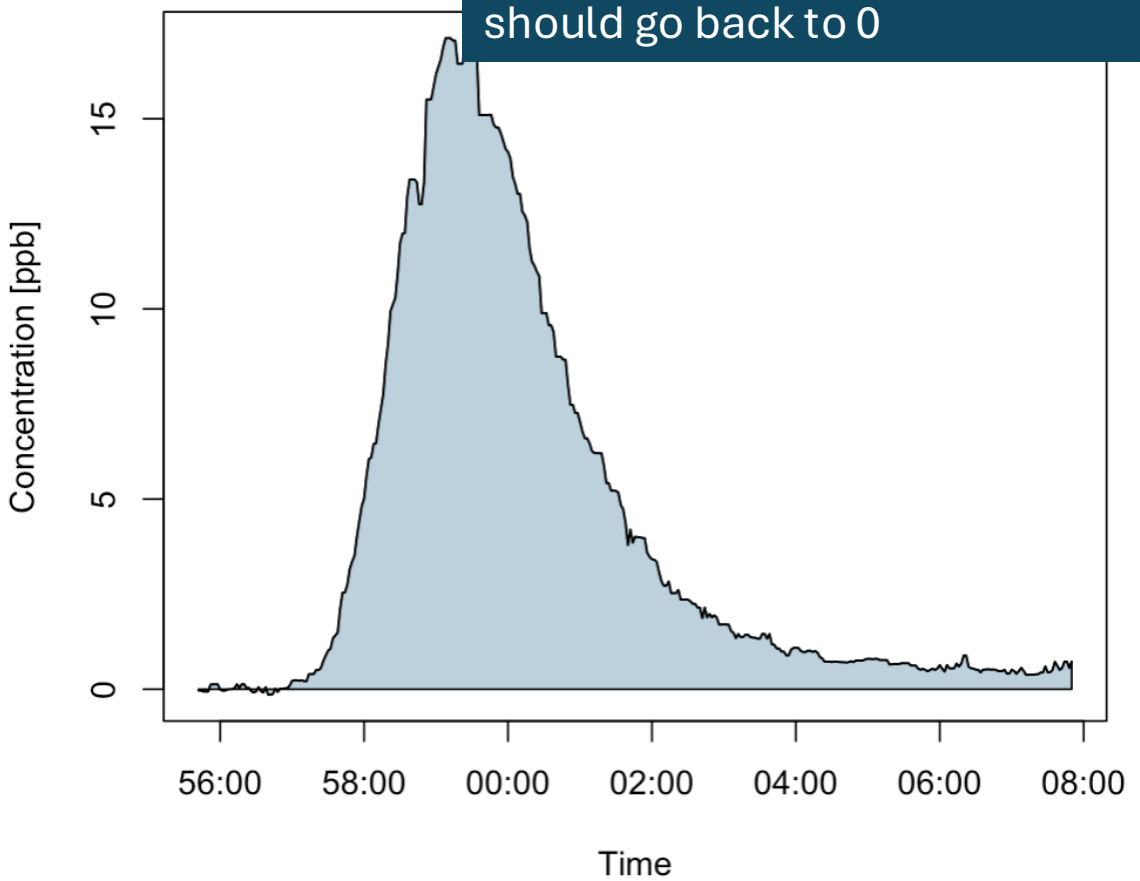
```
> # ---- Curve 1
There were 12 warnings (use warnings() to see them)
> calc_discharge(d)
[1] "Select start point (1/2) in the plot (click) of breakthrough curve ... Press [Esc] to interrupt!"
[1] "Select end point (2/2) in the plot (click) of breakthrough curve ... Press [Esc] to interrupt!"
[1] "Parameters: Tracer mass=6000 mg  ; Standard=50 g/l  ;  Tracer volume=0.02 l"
[1] "Calculated discharge is: 2.00116 m^3/s"
[1] "Calculated discharge (with extrapolation) is: 1.83111 m^3/s"
>
```

The result with extrapolation extrapolates the curve:
More color -> lower discharge





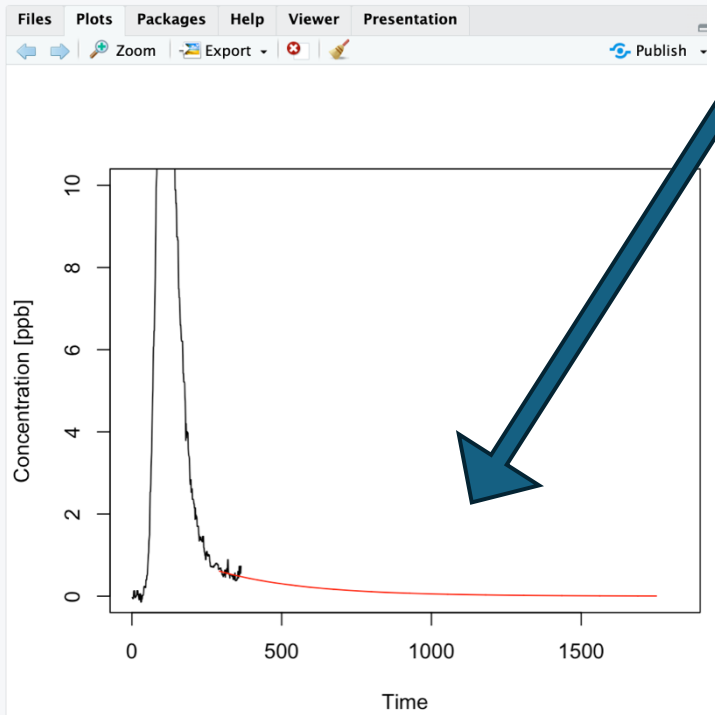
The “**breakthrough curve**” should go back to 0



Action: “Calculate Discharge”

```
> # ---- Curve 1
There were 12 warnings (use warnings() to see them)
> calc_discharge(d)
[1] "Select start point (1/2) in the plot (click) of breakthrough curve ... Press [Esc] to interrupt!"
[1] "Select end point (2/2) in the plot (click) of breakthrough curve ... Press [Esc] to interrupt!"
[1] "Parameters: Tracer mass=6000 mg  ; Standard=50 g/l  ;  Tracer volume=0.02 l"
[1] "Calculated discharge is: 2.00116 m^3/s"
[1] "Calculated discharge (with extrapolation) is: 1.83111 m^3/s"
>
```

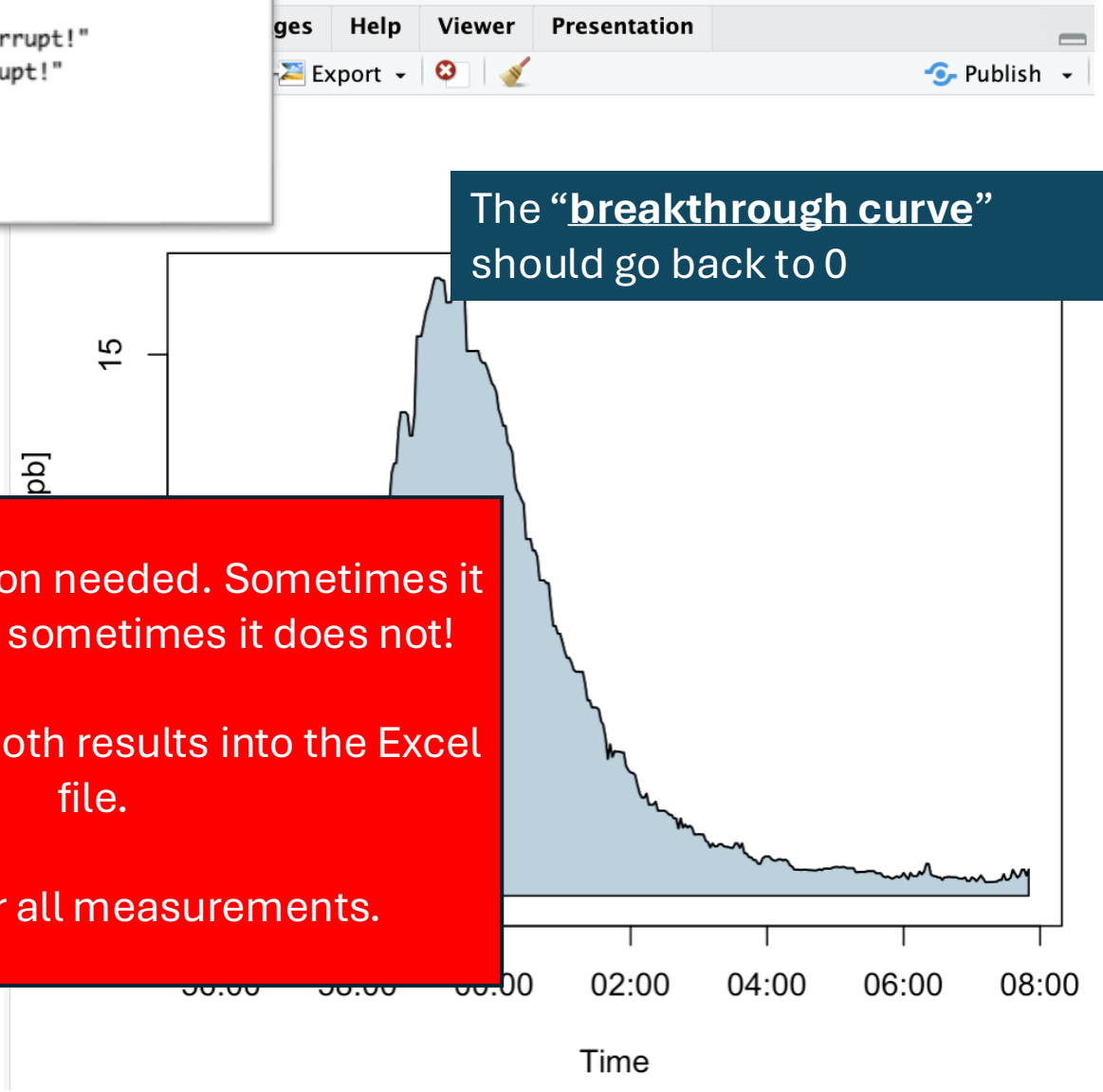
The result with extrapolation extrapolates the curve:
More color -> lower discharge



Visual inspection needed. Sometimes it makes sense, sometimes it does not!

You can write both results into the Excel file.

Repeat for all measurements.



The “**breakthrough curve**” should go back to 0